

Artificial Intelligence and Large Language Models in Advancing Cyber Threat Intelligence: A Systematic Literature Review

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Abstract

The evolution of Cyber Threat Intelligence (CTI) is pivotal in addressing the increasingly sophisticated landscape of cyber threats. Traditional CTI methods, while foundational, are rapidly obsoleting by the complexity and dynamism of modern cyber threats. This demands a shift towards more advanced, adaptive strategies integrating Artificial Intelligence (AI) and Large Language Models (LLMs). This literature survey examines recent developments in CTI, focusing on traditional methods, the incorporation of AI for enhanced forecasting and detection, and the pioneering application of LLMs for automating intelligence report generation, enhancing threat detection, recognition, and mitigation accuracy, and facilitating the construction of comprehensive threat prevention framework. Key findings indicate that while traditional CTI methods provide a critical baseline for threat intelligence, their static nature and reliance on outdated indicators of compromise limit their effectiveness against novel, sophisticated cyber threats. Integrating AI techniques, particularly machine learning and deep learning, marks a significant advancement in CTI, offering improved predictive capabilities, dynamic threat detection, and more nuanced analysis of cyber threats. Further, the advent of LLMs in CTI represents a transformative shift, enabling the automated generation of intelligence reports, processing of unstructured CTI data, threat detection, recognition, and actionable step generation to mitigate potential cyber threats. However, challenges persist, including data privacy concerns, the dynamic nature of cyber threats, and the integration of LLMs into existing cybersecurity frameworks. The reviews provide limitations for each study and future research directions with particular emphasis on developing adaptive, intelligent CTI systems capable of proactively addressing the evolving cyber threat landscape.

INTRODUCTION

The advancement of information and communication technology, epitomized by the Internet, has significantly changed everyday life. The deployment of Internet technology, while facilitating convenience, concurrently increased cyber security vulnerabilities. Recent advancements in IT infrastructure interconnectivity and ongoing innovations in digital technologies are increasing the complexity of systems, inversely affecting system security [1]. This increasing trend in digital systems has catalyzed a transformation in the nature, capabilities, and complexity of cyber-attacks compared to a few years prior, impacting individual users, businesses, and government entities. Modern cyberattacks are not merely the actions of digital activists or online miscreants. These attacks are now orchestrated by highly funded and organized adversaries, transitioning from recreational hacking to sophisticated attacks driven by financial or political motives. This shift has replaced malicious attacks with more covert, adaptive, and persistent strategies. Consequently, terms such as advanced malware and Advanced Persistent Threats (APTs) have emerged. Despite annual expenditures exceeding, organizations are increasingly vulnerable to this new wave of cyber attacks that easily bypass traditional defenses like conventional firewalls, intrusion prevention systems, anti-virus software, and security gateways. These traditional defenses, designed to counter previous attack generations, predominantly rely on static, signature-based, rule-based, or list-based pattern-matching technologies [2].

The frequency of cyber-attacks is increasing globally, posing significant challenges for entities in protecting their informational ecosystems against persistent and proficient adversaries. Cybersecurity protects individuals or organizations' digital data from unauthorized access. Cyber-attacks, which refer to efforts of unauthorized access, often involve the exfiltration of sensitive data, proprietary intellectual assets, confidential business plans, and the compromise of vital information technology infrastructures [3]. There is an increasing trend of organized crime groups and nation-state affiliated cyber units employing cyber-attacks as a tactical maneuver, presenting complex defense challenges for governments and corporate entities despite the development of structured cybersecurity frameworks [4].

Cybersecurity is not just a technological issue; it has become a crucial strategic concern that impacts the entire organization. A PwC survey revealed a significant increase in awareness among CEOs, with 77% identifying cybersecurity as a primary threat to organizational growth, markedly up from 61% in the preceding year. The concept of cybersecurity encompasses strategic decision-making regarding necessary compromises, necessitating a holistic approach and endorsement at the executive level [5].

The World Economic Forum identifies cyber security as a crucial risk associated with substantial financial repercussions [6]. In safeguarding network environments against hostile incursions, implementing security apparatuses, including firewalls and intrusion prevention/detection systems, is a standard procedure. These systems operate on predefined rules, attack signatures, and detection models to either thwart or identify attacks. Nonetheless, a notable limitation of these systems is their reactive nature, often failing to preemptively identify hostile entities until an attack is underway. Additionally, the effectiveness of these systems can be compromised by the intrinsic inaccuracies present in their detection methods. Against this backdrop, Cyber Threat Intelligence (CTI), encompassing detailed knowledge of threats such as malicious IP addresses, domain names, hash values, and the TTP of adversaries, emerges as a critical element in strengthening defense mechanisms [7].

The general workflow of CTI is shown in Fig. 1, which involves collecting, analyzing, and distributing data to detect, identify, forecast, and mitigate cyber threats, enhancing proactive cybersecurity measures. It aids in identifying vulnerabilities before exploitation. CTI enables early detection of threat actors' methods, allowing targeted configuration of intrusion detection systems (IDSs). It's crucial to set these systems based on specific attack patterns and actors analyzed from intelligence data. Additionally, CTI assists in creating robust security plans against cyber threat actors' tactics, essential for effective prevention, detection, and response to cyber threats [8].

Nowadays, network administrators have increasingly leaned on CTI as a strategic tool for combating malicious sources. This includes the application of CTI in configuring firewalls with blacklists to pre-emptively block hostile entities at the early stages of an attack, such as during initial reconnaissance by threat actors. Artificial Intelligence (AI), mainly Machine Learning (ML) and Deep Learning (DL) methods have shown proficiency in pattern recognition and forecasting future actions based on historical data [9]. ML emerges as a prominent technology within the fourth industrial revolution, enabling systems to evolve autonomously and enhance their performance without explicit programming [10].

This advancement in ML has led to its application in cybersecurity, facilitating intelligent analysis of security-related data and offering dynamic and modern protective measures. Leveraging these technologies, an array of intelligent cybersecurity applications can be developed, such as ware identification and zero-day attack prediction, among others, by extracting insights from structured and unstructured data. Recently, there has been a significant surge in cybersecurity measures to counteract diverse forms of cyber threats, including unauthorized access, Denial-of-Service (DoS) attacks, phishing, malware, botnets, spyware, worms, and more. Consequently, modern cyber security applications require sophisticated data analysis tools and frameworks that are adept at extracting meaningful knowledge from data quickly and intelligently. This systematic literature review (SLR) aims to explain a comprehensive perspective on conventional, ML, DL, and LLM-based CTI methods, showing their utility in intelligent data analysis and automation within the cybersecurity domain [10]. These methods are skilled at extracting insights from cybersecurity data, thus encouraging data-driven intelligent decision-making and automation, paving the way for a new generation of cyber-defense strategies that are more proactive than existing models. Therefore, the principal focus of this study is to examine the potential application of various CTI fireworks, ML, DL, and LLM methods for threat detection, recognition, forecasting, and mitigation across different domains within the cyber landscape. A systematic review of the scientific literature from the past five years has shown the impact of progressive procedures

and technological advancements on enhancing the cybersecurity of organizational critical infrastructures, explicitly focusing on the enhancement of CTI [8].

1.1 Objectives and Contributions

The primary aim of this SLR is to analyze the recent developments in CTI methods, techniques, objectives, data handling, and limitations to effectively address the inherent challenges associated with the aggregation of CTI data from multiple heterogeneous sources. The ultimate objective is to establish the effectiveness of the Large Language Models (LLMs) that are capable of processing the structured and unstructured CTI data and reports, advancing the threat detection, recognition, and mitigation process, concurrently enhancing CTI operations in areas encompassing data collection from heterogeneous, filtration, analysis, visualization, and generating actionable steps. Furthermore, the main contribution of this research lies in presenting current state-of-the-art ML and DL-based techniques, mainly focused on LLMs, to enhance the efficacy of CTI forecasting and threat mitigation efforts. The scope of this work is primarily concentrated on the progress of the past 5 years, especially on ML, DL, and LLMs-based CTI methods.

The core objective of this study is to establish a point of reference for scholars and industry professionals with a vested interest in acquiring knowledge, exploring, and developing data-driven automated and intelligent systems within the cybersecurity domain, employing ML and DL methods for CTI enhancement with particular focus on LLM based methods. The principal contributions of this research are listed as follows:

1. Initially, we categorize CTI detection, identification, recognition, and mitigation research into three technological groups: Conventional CTI methods, AI in CTI, and LLMs in CTI processing. This categorization enables researchers to gain insights into the current landscape of CTI methods by comparing and analyzing the various methods and models employed.
2. Secondly, we provide an exhaustive list of state-of-the-art CTI methods, their contribution, the objective of the study, and limitations in CTI forecasting, primarily focused on LLMs. This resource is a valuable reference point for researchers seeking to establish baseline performance measures for their investigations.
3. Analysing the viability and adaptability of various ML, DL, and LLM paradigms across a spectrum of real-world contexts in the cybersecurity domain. These contextual scenarios determine the transformational potential of data-driven intelligent decision-making and automation, signaling a paradigm shift towards proactive cyber defense strategies in contrast to previously employed traditional methods.
4. Lastly, we discuss challenges associated with current state-of-the-art CTI techniques, the pros and cons of different AI methods, and why LLMs can be a game changer for analyzing unstructured CTI. As a result, this review facilitates future research on CTI through LLMs.

1.2 Paper Structure

The rest of this article is structured as follows: Section 2 defines the process for choosing primary research studies for our SLR, research questions, and the search strategy used to find the relevant studies. Section 3 summarizes conventional, AI, and LLM-based CTI methods. Section 4 compares and contrasts studies and discusses current trends to find the research gap. Section 5 analyses challenges and open issues to establish future research directions.

MATERIAL AND METHODS

Our review follows the guidelines established by [8], in combination with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) recommendations for conducting systematic reviews [12]. We carefully executed an extensive SLR employing a rigorous iterative planning, examination, and documentation process. Primary studies were carefully selected by applying predefined keywords to reputable scientific repositories. These keywords were carefully assigned to facilitate the explanation of study outcomes that are relevant to our research objectives. Notably, the search terms 'Cyber Threat Intelligence', 'Cyber threat', and 'Cyber Attack' were

predominantly used to find relevant scholarly contributions from ScienceDirect, Elsevier, MDPI, Institute of Electrical and Electronics Engineering (IEEE) digital library, the Web of Science (WoS), the Association of Computing Machinery (ACM) digital library, and the preprint articles from world-renowned universities and online repositories.

The inclusion criteria stipulated the following prerequisites: 1) The selected paper should encompass the different CTI areas, such as forecasting, detection, recognition, and identification, within the context of cyber threat mitigation. 2) The paper's publication date should fall within the temporal bracket of 2019 to February 2024. 3) All papers were confined to journal articles, conference proceedings, and preprints from world-renowned universities and research centers.

Upon the initial database keyword searches, an initial tally of 294 research works was identified, with 28 duplicates. After careful examination of the titles and keywords of each paper against the established inclusion/ exclusion criteria, it was determined that 87 papers still needed to meet the criteria. Additionally, 120 papers are excluded from the full-text article analysis. Consequently, the corpus of papers available for review was ultimately reduced to 59. Following a comprehensive examination of these 59 articles and applying the stipulated inclusion/exclusion criteria, 41 papers were deemed suitable for inclusion in our study.

Figure 2 presents the workflow of SLR from the total number of studies assessed to the last stage of accepted studies. A rigorous quality assessment was conducted to ensure the extracted data's robustness. This assessment evaluates the completeness and relevance of the data, followed by an examination of the accuracy of the information encapsulated within these studies.

Identifying the studies that met the predefined quality criteria involved a multi-stage selection procedure, as follows: Stage 1: Manuscripts must have been published in peer-reviewed journals or conference proceedings, or preprints of renowned institutions are considered after a thorough examination. Publications such as poster presentations, books, and blogs were intentionally excluded due to concerns regarding their quality. Stage 2: The selected papers had to center on CTI, cyber attack, and cyber security and their impact on organizational performance. Stage 3: The considered papers had to fall within case studies, system applications, or modeling implementations. A quality assessment checklist was employed to evaluate identified research articles systematically.

Subsequently, 13 studies were deemed ineligible for inclusion in the analysis due to their non-conformance with the predetermined criteria outlined in the checklist. The pertinent details and data gleaned from each article were carefully catalogued in a tabular format, encompassing crucial information in categorizing each research article. The temporal distribution of publication years of the selected articles is graphically represented in Fig. 3.

2.1 Research Questions

This manuscript's primary objective is to systematically explore the automation of CTI for threat forecasting, detection, recognition, and identification. We thoroughly analyze a collection of pertinent academic articles to identify

key characteristics. This detailed analysis helps us understand the current state of cybersecurity practices to detect, recognize, forecast, and mitigate potential threats. It also provides insights into the structure and content of studies in CTI. Understanding research trends will undoubtedly benefit future research efforts in CTI forecasting.

In the effort to conduct a thorough literature review, the scope of this manuscript is limited to research articles published from January 2019 to February 2024 within the field of cybersecurity related to CTI forecasting. The study conducted here seeks to address the following research questions:

1. What is the current landscape of CTI automation for threat detection, identification, and recognition through rule-based, match-based, text analysis methods and forecasting methods?
2. What is the current state-of-the-art AI based CTI methods and LLMs' efficacy in comprehending and reasoning about CTI?
3. How impactful are the emerging capabilities of LLMs in various downstream CTI tasks and output actionable steps for mitigation?

2.2 Search Strategy

Online research databases were utilized to find pertinent articles from 2019 to February 2024. Multiple online databases catering to this domain were consulted with IEEE Explore, ACM, MDPI, and ScienceDirect, which stand out as the prominent repositories for publications about CTI technology. Along with these databases, we have also included the preprints from renowned research institutes to include the latest research work in our SLR. The following search queries are used to search and select the research studies for this SLR. ("Cyber Threat Intelligence" OR "Threat Intelligence" OR "CTI automation" OR "Cyber Attack" OR "Cyber Threat" OR "Attack Detection" OR "CTI Forecasting" OR "Cyber Security").

The results obtained from all databases went through a thorough cross-referencing process to identify any redundancies. Subsequently, the results were subject to a thorough filtration process based on their relevance to the field of CTI and cybersecurity, a determination primarily made from the abstracts of the retrieved papers. Exclusionary criteria contain papers authored in languages other than English and those that, despite containing the specified keywords, lacked relevance to CTI detection, identification, recognition, forecasting, or threat mitigation. Moreover, included papers were classified based on their CTI methods, primarily assessed by quantifying keyword and phrase frequencies. Notably, the software VOS viewer tool was employed to facilitate the organization and visualization of the bibliometric data. The corpus of literature in this study was selected from esteemed venues within the domains of CTI Natural Language Processing (NLP), ML, DL, LLM, cybersecurity, and data privacy. These sources are catalogued within prominent electronic scientific repositories: the ACM Digital Library, IEEE, Science Direct, and Web of Science. Moreover, we diligently incorporated scholarly works housed within the preprints of renowned research institutes, as this repository serves as a repository for the latest and most cutting-edge research findings. Once we had carefully defined the list of electronic libraries, we proceeded to extract search queries originating from the core research inquiries and the technical elaborations to retrieve relevant research articles.

To explain this process further, Table 1 carefully lists the precise search terms that we derived from the research inquiries and the technical elaborations, divided into two distinct columns. Within Term 1, the initial seven rows include diverse descriptions related to the task of CTI, while the final two entries pertain to various strategies employed in CTI. Concurrently, Term 2 consists of a collection of specialized terminology. The compilation comprises 18 distinct search terms, all harmoniously combined within the table to formulate search queries for electronic repositories.

Table 1
CTI terms used in the formulation of search queries

Term 1	Term 2
Cyber threat intelligence	Cyber threat forecasting
Cyber security	Cyber threat intelligence/CTI LLM
Cyber threat	Threat Intelligence
Cyber attack	Cyber threat intelligence/CTI forecasting
Cyber threat prediction	Cyber Security LLM
Machine Learning/ML Cyber threat intelligence/CTI	Deep learning/DL/CNN Cyber threat intelligence/CTI
Cyber threat identification	Cyber threat recognition
Cyber threat intelligence/CTI framework	Network Intrusion
Cyber threat intelligence/CTI sharing	Intrusion detection LLM

2.3 Study Selection

We implemented the search terms listed in Table 1 to retrieve 294 research articles. Figure 4 illustrates the keyword distribution of the reviewed articles in the form of a heat map. In this SLR, the majority of search outcomes were obtained from ACM, Science Direct, Web of Science, and IEEE, comprising 208 papers and 58% of the overall count.

It is important to note that a certain degree of redundancy and irrelevance was encountered within the retrieved literature. Consequently, strict inclusion and exclusion criteria were carefully applied to identify the most relevant publications for our review. The following inclusion criteria were employed to select publications for inclusion in this review: 1) Research works employed rule-based, match-based, text analysis, ML, DL, and LLM techniques to detect, recognize, identify, and forecast cyber threats and CTI forecasting. 2) Publications with a publication date from the year 2019 onwards to February 2024. This restriction was imposed to focus on recent advancements in the field of CTI forecasting, given that prior works have been comprehensively surveyed in existing literature. 3) Research papers of substantial length or those published

in the primary track of esteemed conferences, journals, and preprints belonging to world-renowned universities. Conversely, publications that met any of the following exclusion criteria were excluded from our review: 1) Works published before 2019. 2) Publications need more substantial content related to modeling CTI. 3) Studies were conducted on experimental platforms other than desktop computers. 3) Duplicated works.

After carefully applying the aforementioned inclusion and exclusion criteria, we identified and screened 41 research papers. A keyword analysis was conducted to explore the distinct elements of CTI. The resulting heat map shown in Fig. 4 highlights the key terms that emerged from the reviewed articles. This visualization organizes the keywords into color-coded clusters and illustrates their interconnections. Furthermore, it shows the principal topics within this domain by showcasing their frequency in reviewed articles. The heat map effectively reveals the critical terms related to CTI, illustrating the core ideas central to this study area.

LITERATURE REVIEW

The primary investigations revolved around categorizing research topics related to conventional CTI, AI in CTI, and LLM-based, which are discussed more comprehensively according to the specific focal points of individual

publications. Section 3.1 discusses the traditional methods proposed for threat detection, recognition and identification based on task-specific templates, rule-based or NLP techniques. Then, in Section 3.2, AI-based CTI methods encompass research endeavors aimed at enhancing assault detection methodologies, introducing innovative detection models, or leveraging novel detection attributes for threat identification, detection, recognition, and real-attacker identification. Advancing LLMs in CTI, Section 3.3 discusses the attention mechanism-based transformer encoder and decoder LLM models that process unstructured CTI reports for threat detection, type classification, malicious pattern identification, actionable step generation for threat handling and mitigation, and the optimization of decision-making processes through questioning-answering. After the categorization process, a deeper examination was performed to distill the essential elements of each paper. In this section, the various kinds of algorithms, methods, objectives, limitations, and technical details of different CTI systems are discussed in depth. The general acronyms of this systematic review are listed in Table 2.

Table 2
List of Acronyms used in systematic review

Abbreviations	Full Form
AI	Artificial Intelligence
DL	Deep Learning
ML	Machine Learning
FSL	Few-Shot Learning
IoT	Internet of Things
CNN	Convolutional Neural Network
CTM	Cyber Threat Management
RNN	Recurrent Neural Network
LSTM	Long-Short Term Memory
BiLSTM	Bidirectional Long-Short Term Memory
DFR	Digital Forensic Readiness
IOC	Indicators of Compromise
GRU	Gated Recurrent Unit
OSCTI	Open-Source Cyber Threat Intelligence
XAI	Explainable Artificial Intelligence
CRF	Conditional Random Field
SOCs	Security Operations Center
GCN	Graph Convolutional Network
NLP	Natural Language Processing
CNN	Convolutional Neural Network
BERT	Bidirectional Encoder Representations from Transformers
FL	Federated Learning
FLLE	Fixed-Length Language Encoding
LLM	Large Language Model
GPT	Generative Pre-training Transformer
NER	Named Entity Recognition

3.1 Conventional Methods in CTI

The growing volume of data has been exacerbated by the widespread use of security devices and the increasing complexity of information technology. This surge in data has posed challenges for digital forensics and information security regulations. Table 3 summarizes research papers on conventional methods in Cyber Threat Intelligence,

highlighting essential information from the reviewed studies. This includes contributions, techniques, objectives, data sources, and limitations of the studies.

Table 3
Summary of studies related to the Conventional methods in CTI

Method	Contribution	Technique	Objective	Data Source	Limitations
DFR [14]	DFR combines CTI with forensic preparedness, aiming to enhance the state of DFR while concurrently mitigating the temporal and fiscal exigencies associated with incident response and investigative efforts. Implementing actionable CTI is a pivotal mechanism for augmenting the proficiency levels inherent to DFR.	Digital Forensic Readiness model for log analysis	Efficiently identifying and mitigating the primary determinants of security incidents within a limited timeframe.	Audit Log Database	Generate false positives when correlating IoCs from different sources; Lack of zero-day threat detection; Lacks scalability in enterprise-scale environments.
Decentralized CTI [21]	Introduces a decentralized blockchain-based framework protection mechanisms against cyber threats by leveraging distributed data collection and sharing.	A decentralized, blockchain-based CTI-sharing network designed to enhance the security of ISPs, home networks, and IoT devices.	It aims to automate Protection mechanisms by leveraging distributed data sharing of CTI across various stakeholders in real time.	Data collected from Microsoft Azure for the Ethereum at the ISP and the customer premises, equipment routers.	Depends on a limited number of transaction nodes and validators; The gatekeeper component of the framework only examines the information in the OSI layer three.
TIAM [13]	It examines unstructured CTI reports, classifies and identifies cyber threats by leveraging the textual characteristics and subsequently extracts IOC information.	Vector space model that involves text analysis, regular expressions, and threat scoring to assess the quality of CTI	Quality assessment of CTI	APT attack reported by Indian Infosec Consortium	Data sparsity; Lack of Real-Time Assessment; Manual Feature Engineering.
Eight-step CTI [17]	Addressing the challenges associated with collecting, filtering, sharing, visualizing, and analyzing CTI data from various sources with a unified method.	Analytical data insights	Gathering CTI data from diverse sources and enhances the entire CTI process	Pegasus Spyware (2021) and SolarWinds Orion Trojan (2020).	Relies on structured data in STIX/TAXII format; Lack of Real-time Threat Intelligence.

Method	Contribution	Technique	Objective	Data Source	Limitations
AttackKG [18]	Developed a template-based method to construct attack graphs for CTI report parsing to identify attack techniques with templates.	Technique Templates and Graph Alignment	Automatic extraction of structured graphs of attack behaviors from CTI reports, to identify the relevant attack techniques.	1,515 real-world CTI reports obtained from Cisco Talos Intelligence Group and Microsoft Security Intelligence Center	Templates are well structured, while CTI reports are inherently unstructured; Limited ability for the dynamic and evolving nature of cyber threats.
SECDFAN [19]	Developed a semantic schema for generating and sharing of CTI in a standardized format, utilizing raw CTI data sourced from discussion forums.	Multi-layered system consists of selection, surveillance, processing, analytics, presentation, communication, and collaboration layer	Leveraging discussion forums as a source of raw CTI data	Data from discussion forums across the surface web, deepnet, and darknet	CTI products generated by SECDFAN could be impacted by the dynamic and complex nature of cyber threats; heavily depends on the NLP techniques.
Cyber Defenders [20]	This study aims to methodically identify false-positive occurrences across all services and leverage this data to recommend areas for improvement. Additionally, it addresses the recognition and documentation of challenges encountered in the detection and analysis of vulnerabilities, as well as the integration of an inventory source into the SOC for ongoing monitoring.	CTI lifecycle framework identifies, categorizes, and addresses SOC vulnerabilities to optimize cyber defense and risk management.	Address false positives, error states, and operational problems encountered in the detection, analysis, and remediation of vulnerabilities within SOCs	SOCs data focusing on the classification of false positives, error states, and operational issues in vulnerability detection and remediation.	It operates under the assumption of a relatively uniform threat landscape, which may not hold true for organizations facing highly specialized or targeted threats; One-size-fits-all approach may not be practical for every organization, especially those with unique risk profiles or those operating in highly dynamic environments.

Method	Contribution	Technique	Objective	Data Source	Limitations
BLOCIS [22]	Developed a blockchain-based open CTI framework for enhancing data validity with traceability, integrity, and Sybil-resistance features. It involves collecting and sharing threat data, consumers for utilizing this data, and feeds for CTI data sharing to thwart Sybil attacks from malicious contributors.	Blockchain-based smart contracts	Address the threat of malicious contributors and miners who may attempt to inject false data or exploit the system for unwarranted gains.	Simulations for Sybil attack	The framework's security and reliability heavily rely on the underlying smart contracts. Attackers could exploit vulnerabilities or flaws within these contracts. Storing a large amount of CTI data directly on the blockchain can lead to blockchain bloat.
CyRiPred [16]	Detection of primary cyber risk cohorts from an exhaustive repository of security vulnerabilities is a pivotal endeavor and future risk assessment	Statistical cyber risk recognition and prediction method	Data-driven attack forecasting	National Vulnerability Database	Lack of continual learning; Lack of Zero-Day Detection.
ABC [15]	Proposed a secure CTI sharing, categorization, and PoQ	Blockchain-based probabilistic model	Blockchain-based secure collection, evaluation, storage, and sharing of CTI	Simulation-based CTI-feeds	Biases due to evaluation criteria; Heavily relies upon the reputation of validators and CTI Feed Sources.
SCERM [50]	Proposed a CTI framework for the identification of cyber threats across various stages, and components of threat intelligence, along with their interconnections and dependencies.	SCERM is based on the STIX framework, which has three main modules: Preprocessing: Parses STIX reports to extract and store relevant Cyber Threat Intelligence (CTI) in a graph database. Valuation: Assesses STIX data to generate scores for prevention, detection, and	Valuation of the STIX reports and completion of missing content related to various phases of CTMt.	Schema test, HAILATAXII, and IBM X-Force Exchange	Potential vulnerability to low-integrity threat sources that could mislead by reporting false threat data; SCERM does not verify the accuracy of the provided CTI data, assuming sources are trustworthy.

Method	Contribution	Technique	Objective	Data Source	Limitations
		response. Refinement: Identifies and prepares incomplete components within the STIX data.			

Digital Forensic Readiness (DFR) [14] is introduced to enhance security incident response. This model integrates CTI with DFR, reducing investigation time and costs. It primarily involves IoC collection, audit log processing, and threat identification. Through experiments, it demonstrates the model’s effectiveness in identifying malicious activity, which showed a high accuracy rate in identifying malware instances, significantly reducing the volume of data for analysis and improving forensic investigation efficiency. This CTI-informed DFR model can enhance an organization’s security posture and incident response capabilities. Security is a major issue for the setup and growth of the Internet of Things (IoT), highlighted by recent cyberattacks on important Internet service providers. These attacks brought down large parts of the internet by using weaknesses in unprotected, embedded devices, often found in home networks managed by users without knowledge of cybersecurity best practices. Therefore, creating security solutions is critical to protect against risks in insecure, untrusted public networks by implementing automated defenses and enabling easy, affordable sharing of network security information.

A decentralized blockchain technology framework is proposed to enhance cybersecurity across Internet Service Providers (ISPs), home networks, and IoT devices [21]. The framework aims to automate the sharing of CTI in near real-time through a consortium blockchain network, addressing the challenge of securing IoT devices against cyber threats without centralized trust. This approach leverages blockchain’s inherent properties, such as immutability, decentralization, and secure exchange of digital assets, enabling smart contracts among devices for autonomous interaction and proactive and reactive cybersecurity measures. The proposed system collects first-hand CTI from network participants, including end-users, network components, and detection systems, and shares this information securely among stakeholders. It differentiates between two tiers: one for end-users and their service providers for privacy and protection from external threats and another for service providers to exchange CTI without direct end-user involvement. It shows the feasibility of using blockchain to secure network infrastructures and IoT devices with minimal network architectural changes, offering a scalable and sustainable solution.

Automated Assessment Model TIAM [13] was proposed for the automatic quality assessment of the unstructured CTI reports. TIAM engages in the CTI content analysis by extracting Indicators of Compromise (IOCs) in vector space format. These IOCs facilitate the evaluation of CTI’s value and empower security managers to identify compromised targets. Notable IOCs encompass IP addresses, domain names, malicious file hashes linked to Command and Control Servers (C2 servers), malware downloads, and other pertinent characteristics. Throughout the CTI assessment process, TIAM is harmoniously integrated with the ATT&CK matrix, aiding security managers in examining potential attack techniques employed by adversaries and devising corresponding mitigation strategies.

Significant increase in cyber attacks across various sectors, highlighting the challenges organizations face due to the vast amount of heterogeneous threat data generated from numerous sources that need to be efficiently ingested, processed, and analyzed to derive actionable insights for threat mitigation. The proposed framework consists of an eight-step model for CTI, emphasizing the timeline visualization of threats and analytic data insights [17]. It includes steps for managing user interactions, indexing and storing data, collecting external data, generating normalized data, searching through the data, sharing information, and finally, visualizing and analyzing the data. This approach aims to

shorten the threat mitigation timeline and improve the overall effectiveness of CTI by ensuring data from various sources can be collected, filtered, shared, and analyzed coherently and structured. A key aspect of the proposed framework is its focus on improving the understanding of the context in which threats occur, thereby making mitigating vulnerabilities more effective. It involves developing a tool that allows security analysts to input threat data, analyze this data, and create a timeline visualization that offers a clearer understanding of threats and their progression. Including visualization and analysis as the last two steps highlights the emphasis on collecting and sharing data, deriving meaningful insights through analytical processes, and presenting the data in an easily interpretable format [17].

The rise in complex and varied cyber attacks poses a significant challenge to identifying and stopping these threats. Security experts share their insights on these threats through CTI reports. The proposed template-based framework, AttackKG, turns the text from CTI reports into structured graphs showing how attacks happen and what techniques are used [18]. AttackKG also combines information from different reports to create a complete picture of attack methods and employs templates for aggregating threat intelligence across reports, enhancing the granularity of attack representations. These templates, initialized with MITRE-derived data and refined with insights from CTI analyses, serve as a blueprint for aligning with the extracted attack graphs. A fine-grained graph alignment algorithm underpins this process, enabling the precise identification of attack techniques and their subsequent incorporation into Technique Knowledge Graphs (TKGs). The framework tackles the challenges of parsing CTI reports, such as domain-specific term identification, entity and dependency extraction, co-reference resolution, and graph construction and simplification, through advanced NLP techniques and domain knowledge application. AttackKG is tested by over 1,500 real CTI reports and validated. AttackKG accurately maps out attacks and gathers detailed information on attack techniques from many reports.

The SECDFAN system is a platform for collecting and sharing Cyber Threat Intelligence (CTI), which utilizes discussion forums as its primary data source. Its goal is to enhance the capabilities of information security experts by providing a structured approach to creating actionable CTI products [19]. By focusing on the often underutilized potential of discussion forums containing extensive cybersecurity discussions, SECDFAN effectively analyzes these forums to gather relevant data. A key innovation of the system is its semantic schema, which structures forum data, facilitating intelligence extraction from diverse and informal content. The architecture of SECDFAN follows a systematic method that addresses all stages of CTI, from data sourcing and collection to the distribution of CTI products and collaboration among experts. It includes specific layers for various tasks such as data collection, analysis, presentation, and quality control, transforming raw data into standardized CTI products. SECDFAN employs Natural Language Processing (NLP), semantic analysis, and Machine Learning (ML) to identify cyber threats accurately. It aims to overcome the limitations of current CTI systems by providing a comprehensive framework for leveraging forum data in the generation of CTI. The development of SECDFAN focuses on implementing this architecture and enhancing its capabilities for automatic data annotation and improving the quality of CTI products and data [19].

Vulnerability and threat assessments often assume success while neglecting to document systematic failures. This oversight conceals potential risks and negatively impacts risk management strategies. The relevance of vulnerabilities, detection errors, and delays in remediation all contribute to these issues. Comprehensive documentation, particularly of Security Operations Centers (SOCs) detection failures, enhances visibility into hidden risks and influences strategic decision-making. Rigid categorization aims to align mitigation outcomes with business risk management by identifying intrinsic and extrinsic risks. Systematic identification and management of false positives, error states, and operational challenges are crucial in the field of cyber defense, with specific attention to SOCs highlighted by [20]. This approach underscores the importance of categorizing and documenting these

challenges to facilitate process improvement and strategic risk management. By mapping practical SOC outcome failures into clear risk management categories, this method aims to optimize workflows, improve cyber risk management strategies, and provide valuable metrics for assessing the effectiveness of cyber defense mechanisms. The proposed Cyber Threat Intelligence (CTI) lifecycle divides vulnerability management into three stages: detection and relevance rating, timing for activation of prevention measures, and detection of exploitation signs. It also suggests integrating dimensions of business processes (people, products, and partners) into resolution category calculations, thereby linking cybersecurity directly to risk quantification for effective and efficient management. This approach advocates for incorporating feedback mechanisms into existing processes to optimize workflows and provide valuable metrics [20].

The integration of 5G and IoT technologies has significantly increased the diversity and complexity of network environments, resulting in a wider range of attack vectors for cyber threats. Traditional security systems are inadequate for combating advanced threats such as zero-day attacks and Advanced Persistent Threats (APTs), which exploit unknown vulnerabilities and target specific entities over prolonged periods. While current Cyber Threat Intelligence (CTI) systems are proactive, they often suffer from data integrity and reliability issues due to the potential for attackers to inject malicious data, especially through Sybil attacks. To address this, the Blockchain-Based CTI Sharing Framework (BLOCIS) has been specifically designed to resist Sybil attacks, as outlined by [22]. This innovative framework utilizes blockchain technology to enhance the reliability and validity of CTI data greatly. By implementing smart contracts for the validation process and recording data metadata on a blockchain network, the system ensures the data source's and its contents' traceability and immutability. This approach is crucial for maintaining data validity and reliability, essential for effective CTI operations. BLOCIS consists of contributors who collect and share threat-related data, consumers who utilize this data, and feeds that facilitate CTI data sharing by evaluating the validity of the data before disseminating actionable intelligence. A notable feature of the system is its ability to mitigate the impact of Sybil attacks through a mechanism that assesses the reliability of data contributors. Contributors are rewarded based on the validity of their submissions, as determined by smart contracts that adjust their reliability scores according to the CTI feed's criteria. This model ensures the integrity and reliability of shared data and enhances the network's collective capacity to proactively and effectively respond to cyber threats.

The need to share and promptly respond to high-quality Cyber Threat Intelligence (CTI) with the appropriate stakeholders is crucial. This study [15] developed a probabilistic model based on blockchain technology to collect, evaluate, store, and share CTI while assessing its quality according to predefined standards. This system selects validators and evaluates CTI inputs using a reputation and trust-based method. Quality assessment is achieved through a trust-based validator selection process, where validators rate CTI feeds against predefined standards using a proof-of-quality (PoQ) consensus mechanism. The data are securely stored in a ledger, including clear evaluations and validator performance information. The system's reliability, consistency, and resilience against malicious activities were analyzed theoretically through probabilistic simulations, demonstrating its capability to effectively handle adversarial validators.

The cyber risk posed by malicious external threat actors is a significant socioeconomic concern. Anticipating such risks is a challenging task, made even more difficult by the constantly evolving nature of attack vectors. This investigation introduces a predictive model designed to automate the assessment of cyber risks [16]. The model relies solely on Common Vulnerabilities and Exposures (CVE) data to reduce the influence of subjective expert opinions in forecasting cyber risk. First, the model categorizes CVE data into clusters of cyber risk by aligning the textual descriptions in the database with relevant Wikipedia article titles. After this categorization, CyRiP Red consolidates the incidence and severity of the identified topics over a specified time interval, generating a time series

sequence for predictive analysis. Risk assessments are calculated by combining forecasted occurrence and impact metrics. Finally, the resulting cyber risks are ranked based on their composite scores.

Cyber Threat Management (CTM) involves critical steps for preventing, detecting, and responding to cyberattacks. This process relies on accurate threat identification and the implementation of appropriate actions. However, organizations often face challenges in executing these processes effectively, particularly within the limited time available before an attack occurs. The Structured Threat Information eXpression (STIX) initiative [50] was developed to improve the organization of Cyber Threat Intelligence (CTI) for better sharing, visualization, and analysis. Unfortunately, the quality of publicly available STIX reports is frequently compromised due to poor formatting, inaccurate terminology, and missing or mislabeled critical components, which can undermine their effectiveness in cyber threat management. To address these challenges, the SCERM (Structured Cybersecurity Event Report Management) has been proposed to improve the usability and effectiveness of STIX reports for CTM. This framework tackles common problems such as incomplete, inappropriate, or redundantly placed CTI data. SCERM is structured into three main modules: Preprocessing, Valuation, and Refinement, each comprising various submodules that perform specific functions. The Preprocessing module consists of the Parser and Booster submodules. It processes STIX reports by extracting CTI components and organizing them within a graph database. The Parser is responsible for ingesting data and performing initial extraction, while the Booster refines this data by segregating distinct elements into a Distinct Component List (DCL) and properly categorizing misplaced items into a Native Component List (NCL), ensuring optimal relevance to the phases of cyber threat management. Next, the Valuation module assesses the alignment of STIX data with the prevention, detection, and response phases of cyber threat management. This module generates valuation scores that gauge the efficacy and applicability of the report's CTI. In the Refinement stage, the Component Analyzer identifies deficiencies within the report. The Crawler addresses these deficiencies by sourcing missing or additional CTI from curated datasets. The Adder then incorporates this newly sourced data into the report, enhancing its completeness and utility for CTM. This process is iterative, featuring a feedback loop that ensures continuous improvement until a predetermined valuation score is achieved or improvements plateau. The refined STIX reports, along with their valuation scores, empower analysts to prioritize more effectively and deploy CTI for enhanced CTM [50].

To summarize, Fig. 5 illustrates the general workflow of conventional CTI methods, which is utilized for various downstream tasks. Each phase of this workflow includes multiple methods and techniques, with each CTI method integrating these options.

3.2 AI Methods in CTI

By leveraging AI, CTI frameworks can process large amounts of data and identify complex patterns of malicious activity, providing valuable insights for proactive defense strategies. However, current CTI techniques often struggle to detect new and previously unknown threats, highlighting a significant limitation in the rapidly evolving cybersecurity landscape.

Table 4 summarizes research papers on AI methods in Cyber Threat Intelligence, highlighting essential information from the reviewed studies. This includes contributions, techniques, objectives, data sources, and limitations of the studies.

Table 4
Summary of the AI methods in CTI

Method	Contribution	Technique	Objective	Data Source	Limitations
TIMiner [31]	Proposes a CNN-based system for categorizing CTI across key domains. It also presents the Threat-Index, a novel metric.	Word2Vec for word embeddings, CNN recognizer to identify which CTI belongs to, BiLSTM to recognize IOCs.	Identification of unseen types of Indicator of Compromise (IOC).	Threat data collected from security blogs, social media feed, security vendor bulletins, and posts published in hacking forums.	Dependence on Social Media Data and the possibility of misinformation or incomplete data could lead to skewed threat analysis. Evaluation Against APTs does not experiment.
CTI-Twitter [32]	CTI from Twitter to identify knowledge oriented toward threats within an extensive and swiftly expanding volume of content on Twitter.	Term Frequency-Invert Document Frequency (TF-IDF) and BERT for Multiclassification, Latent Dirichlet Allocation (LDA) clustering for Semantic Grouping.	Relevance extraction and classification to distinguish relevantly threat tweets from irrelevant ones using a fine-tuned pre-trained classifier.	Twitter feed, National vulnerability database, and CERT alerts.	Semantic clustering is not optimized for the short-text format of tweets. CTI-Twitter does not prioritize the information based on its immediacy or relevance over time.
HinCTI [34]	Proposed a HIN with an MIIS measure in a GCN for CTI modeling for enabling the semantic integration of CTI	GCN to identify the threat types.	Threat type detection of cyberinfrastructure nodes.	Real-world data from IBM X-Force Exchange Platform and VirusTotal comprised of 11,340 nodes.	HinCTI analysis includes only four distinct infrastructure node types and five specific relation types. Incorporation of the dynamic threat landscape associated with infrastructure nodes is neglected.
INTIME [30]	Developed a CTI lifecycle for gathering, processing, analyzing, and sharing via integrated ML techniques.	SVM-based SMILE page classifier trained on positive and negative examples of webpages; CNN for tweet classification.	Automated ranking of content based on its potential relevance to CTI.	CTI data from social media, deep, and dark web.	Limitations include zero-day attack vulnerabilities; Limited to social media feed.

Method	Contribution	Technique	Objective	Data Source	Limitations
CTI View [37]	Developed a transformer-based threat entity identifier and to accurately extract threat entities from CTI text that successfully identify attackers, exploits, regions, industries, and campaigns described in the texts.	BERT to analyze security texts on APT events with an added GRU layer for entity extraction.	Development of a novel automation CTI system, CTI View for extracting and analyzing vast amounts of unstructured CTI shared by diverse security vendors.	English APT threat Intelligence corpora.	The proposed CTI View system could achieve 72% accuracy only; The system focuses on extracting entities related to APT threats but does not focus deeply on relation extraction between entities.
Malware forensics [28]	A novel tri-layer CTI framework that separates the Threat feeds collection, Semi-supervised analysis, and threat mitigation.	Ensembled Xgboost and random forest for classification.	CTI report filtering into the malicious and non-malicious.	Sandboxed environment, Malimg, and Microsoft's Malware Dataset.	Misclassifications with zero-day attacks; Complex layered pipeline.
THREATKG [29]	Develops a novel OSCTI gathering and management, leveraging ML and NLP to extract and structure threat intelligence into a knowledge graph, continuously updated with new data.	ML classifiers for filtering OSCTI reports, BiLSTM for NER, PCNN-ATT for RE, and Threat Knowledge Graph.	Measuring the effectiveness of THREATKG in extracting the threat knowledge entities and relations for classifying the malicious and non-malicious OSCTI reports.	OSCTI/reports from various sources.	OSCTI sources might be adversarial or compromised; The modular structure of THREATKG lacks end-to-end gradient passing; The schema might not be able to capture novel information.
TriCTI [33]	Developed an automatic actionable CTI discovery system by processing cybersecurity reports as input and selects sentences containing potential IOCs	TriCTI classification based on elaborate trigger vector for expressing the campaign stage and weight improvement.	Enhance accuracy of actionable CTI as compared to BERT.	2,362 and 805 sentences in DS-1 and DS-2, respectively. A total of 3,468 IOCs in DS-1 and DS-2.	Sentences describing cybersecurity attacks may encompass multiple campaign stages, leading to contradictions within the model; TriCTI misaligns IOCs with stages by misinterpreting

Method	Contribution	Technique	Objective	Data Source	Limitations
	and classifies their campaign stage.				sentences and technical terms via regular expressions.
Botnet Detection [35]	It introduces computable CTI to reduce human input in cybersecurity decisions, leveraging Explainable AI (XAI) and OSINT are used to improve ML model transparency, as exemplified through DGA detection. Additionally, it presents a model for botnet DGA traffic detection that excels with only seven features, outperforming previous methods and character-based deep learning models against sophisticated DGA attacks.	Botnet DGA detection based on Random Forest, chosen for its effectiveness in handling high-dimensional features.	Enhancing the detection of malicious traffic by employing a blend of XAI and OSINT to improve the trustworthiness and explainability of CTI sharing.	Malicious DNS data collected using Alexa Top.	Computational cost to calculate certain features, Specifically Min-RE-Botnets and IRad, is highlighted as a limitation due to their linear time complexity; limited robustness to generalize across different types of DGA adversarial attacks.
BFLS [36]	Introduces a novel CTI sharing method combining Federated Learning (FL) and blockchain (BFLS) to secure CTI exchange. It enhances the blockchain's consensus protocol to select high-quality CTIs	Blockchain and FL-based K nearest neighbor algorithm	Decentralized training of threat detection models and using blockchain technology for the secure, decentralized aggregation and sharing of these models, BFLS aims to mitigate privacy concerns and security risks associated with traditional CTI sharing methods.	ISCX-IDS-2012 and CIC-DDoS-2019 datasets have 13 different types of DDoS attack	BFLS uses a CTI validation committee to remove bad or harmful CTI. However, its ability to protect against advanced attackers or cooperation among harmful nodes is not fully explored. Concerns exist about maintaining over 50% control, potentially undermining the

Method	Contribution	Technique	Objective	Data Source	Limitations
	for FL integration, with models automatically updated through smart contracts.				committee's effectiveness.
CTI attribution [24]	Development of a CTA framework for detailed and accurate profiling of cyber-threat actors to identify real individuals or organizations behind attack	Attack2vect based on word2vec to extract features from unstructured CTI reports; Random forest for CTA attribution.	CTA to know about the actual attacker (individual or organization) behind the attack.	CTI reports and Malware Sample reports gathered from multiple data sources	Limited availability of CTA dataset data sets; Lack of standardized report format for unstructured reports complicates the extraction of useful information.
Social Media-CNN [25]	A novel approach for identifying, analyzing, and reporting a cyber threat index for various countries based on social media content. Also, CNN-based anomaly detection to identify and report global cyber threat anomalies.	CNN-based anomaly detection	Cyber threat detection from Twitter feed data and reporting a and new cyber threat indexes for various countries	Twitter feed	Reliance on social media data for generating cyber threat indexes makes it vulnerable to any disruptions or manipulations within these platforms; Susceptible to information operations, where large volumes of fake posts related to cyber threats are generated.
Intrusion Detection [26]	A novel privacy-preserving CTI scheme using Federated Learning for cyber attack detection	LSTM with Federated Learning	Intrusion identification in the network while keeping the confidentiality, integrity, and availability of the network data	NF-UNSW-NB15- v2 and NF-BoT-IoT-v2	Inefficiency in processing highly heterogeneous data sets; domain adaptation requires to address statistical discrepancies
Few-shot CTI [27]	Continuous learning CTI classifier with a small number of training instances through few-shot learning	Few-shot learning and transform learning based BERT	Enhance the generalizability of the new specialized CTI classifiers for new cybersecurity events, such as new malware names or novel attack contexts	Microsoft Exchange Server data breach of 2021	Experiment is limited to the BERT base model.

Method	Contribution	Technique	Objective	Data Source	Limitations
CTIMD [23]	Proposed a dynamic malware detection model that combines threat data and API call sequences	TextCNN and BiLSTM	Capturing the interrelationships between parameter values and hostile behaviors	Datacon 2019 malicious code dataSet	It does not address the limitations of static detection techniques, which can be easily evaded by obfuscation techniques and are ineffective in detecting unknown; Limited generalization with heuristic rules
Deep Maxout [51]	Leverages DL and blockchain for threat intelligence and situational awareness. It employs an entropy weighting power k means for alert segregation for identifying potential security threats and DMN for predictions of potential intrusions	Blockchain with ATS for alert segregation and modified-DMN for future alert prediction	Blockchain-based cyber intrusion detection and mitigation	Mendeley dataset	Limited ability to handle zero-day attacks; The proposed system only works with structured data.
CyberEntRel [52]	Developed a novel joint extraction technique utilizing an attention mechanism for the identification of cyber entities and their relationships within textual data. This method combines sequential	Transformer-based RoBERTa model to create vector representations of input words. BiGRU for processing embedding vectors to calculate a probability distribution. Conditional Random Field (CRF) layer for determining the most likely sequence of tags	DL-based novel technique development for extraction of cyber entities and relations from textual information.	1098 CTI reports gathered from various sources such as Microsoft, Cisco, McAfee, and Kaspersky.	Decrease in precision and recall when extracting Overlapping relational triples compared to non-overlapping ones; Does not address new and zero-day threats.

Method	Contribution	Technique	Objective	Data Source	Limitations
	tagging with relation-matching techniques to derive relation triples from CTI data efficiently.	across the entire input, incorporating tag dependencies to ensure the output constitutes a coherent sequence of entities and relations.			

Cybersecurity teams are increasingly adopting Cyber Threat Intelligence (CTI) sharing as a strategic approach to enhance defenses against cyberattacks. However, the effectiveness of these initiatives is questionable due to two main shortcomings: first, existing methods struggle to detect previously unknown Indicators of Compromise (IOCs), and second, they fail to automatically produce categorized CTIs with domain-specific tags (e.g., finance, government), which makes CTI sharing less effective. To address these issues, TIMiner has been proposed for the extraction and sharing of CTI, utilizing data from social media platforms, as noted by [31]. TIMiner employs a convolutional neural network (CNN)-based domain recognizer to assign CTIs to their respective affected domains. Experiments conducted on over 50,000 security-related texts focus on five particularly vulnerable domains: finance, government, education, Internet of Things (IoT), and Industrial Control Systems (ICS), achieving an accuracy rate of 84%. Additionally, TIMiner introduces a novel automated method for extracting IOCs using word embeddings and syntactic dependencies, which has achieved 94% accuracy and 92% recall, resulting in the extraction of over 1.28 million IOCs from security texts. It also presents the Threat Index, a metric for assessing the security levels across different domains by analyzing the impact and severity of threats. Moreover, it includes a detailed analysis of threat evolution based on over 118,000 texts and posts from January 2002 to November 2018 [31].

As cybercrime continues to rise, organizations encounter difficulties in acquiring timely and relevant cybersecurity threat intelligence due to the fragmented nature of knowledge across various platforms. CTI-Twitter addresses this issue by utilizing both supervised and unsupervised machine learning algorithms to extract, process, and generate targeted cyber threat intelligence from Twitter, which serves as a rich source of diverse threat data [32]. CTI-Twitter starts by collecting data from Twitter using its API and preprocesses it into a structured format suitable for analysis. It then employs a refined pretrained classifier to detect tweets that contain security-related information, categorizing them into five specific classes: INFO, ATK, MAL, CVE, and MISC. Semantic clustering is applied to these categories to identify closely related tweets and trending keywords, which enhances data enrichment and facilitates targeted intelligence collection.

Cyberattacks are increasingly complex and organized, prompting a greater emphasis on Cyber Threat Intelligence (CTI) for better understanding and mitigation of cyber threats. However, modeling CTI is challenging due to the intricate connections between various elements and the lack of labeled data for infrastructure nodes, complicating automatic threat identification. To address this, HinCTI was developed [34] for threat type identification. It employs a threat intelligence meta-schema to semantically associate infrastructure nodes, utilizes a Heterogeneous Information Network (HIN) for integration, and incorporates a Meta-path and Metagraph Instances-based Similarity (MIIS) measure alongside a heterogeneous Graph Convolutional Network (GCN) for effective identification. HinCTI enhances the integration of nodes such as domain names and malware hashes while employing hierarchical regularization to minimize overfitting, aiming for improved accuracy despite data constraints. Currently, it focuses on four node types and five relationship types but plans for future expansion to include more entities. A current limitation is its static approach to threat types, which does not account for evolving threats.

The widespread use of technology across various devices has created significant cybersecurity challenges due to diverse operating systems and platforms. Focusing on Cyber Threat Intelligence (CTI) is essential to combat evolving cyber threats. CTI involves gathering, analyzing, and sharing data to prevent or respond quickly to cyber incidents. INTIME is a machine learning framework designed to identify, extract, and distribute CTI from sources like the dark web, forums, and social media [30]. It utilizes automated natural language processing for tasks such as entity disambiguation and supports robust CTI management. The data acquisition module gathers cybersecurity data using web crawlers and social media monitors, employing algorithms like Random Forest and Convolutional Neural Networks (CNN) to classify this information. The analysis module processes the data with techniques like Word2Vec for threat identification, while a Custom Platform Enumeration (CPE) suggestion engine enhances data linkage. Management and sharing of CTI are supported through MISP scripts, creating a structured intelligence repository. Experimental evaluations demonstrate the effectiveness of the Random Forest classifier and emphasize the importance of precise keyword selection in social media monitoring. This framework aims to improve security analysts' ability to identify and respond to cyber threats effectively.

The CTI-View system is designed for Cyber Threat Intelligence (CTI) analysis, mainly focusing on Advanced Persistent Threats (APTs) through an innovative application of Natural Language Processing (NLP). It features a new text extraction framework that integrates automated testing, text recognition, and denoising techniques to process unstructured CTI data from various sources efficiently. This approach effectively addresses challenges such as non-standard HTML and anti-crawler mechanisms. The system excels at extracting Indicators of Compromise (IOC) and Tactics, Techniques, and Procedures (TTP) by employing a combination of regular expressions along with a refined blacklist and whitelist mechanism. Central to CTI-View is an advanced entity extraction model based on BERT, supplemented with a Gated Recurrent Unit (GRU) layer, significantly improving entity identification accuracy [37]. This model has been thoroughly evaluated against a dataset annotated with cybersecurity-related entities, showing superior performance compared to traditional models. CTI-View marks a significant advancement in cybersecurity threat intelligence, providing automated and accurate complex data analysis to support proactive defenses against APTs. The system focuses on extracting and analyzing the large volumes of unstructured CTI supplied by various security vendors.

The increase in cybercrimes has prompted organizations to shift from reactive to proactive cyber defense strategies. The proposed Tri-Layered Cyber Threat Intelligence (CTI) framework consists of three distinct layers: data management, filtering, and reporting, as noted by [28]. The first layer, Layer 1, focuses on the collection and storage of data. It utilizes both malicious and non-malicious sources, including open-source threat feeds, sandbox environments like Dionaea, and various benchmark datasets. This approach ensures a robust and comprehensive foundation for subsequent analysis. Moving to Layer 2, this stage is crucial for monitoring and analyzing the behavior of identified threats. It employs advanced techniques such as Feature Engineering, Gradient Boosting (XGBoost), Ensembling, and Semi-Supervised Learning. Cross-validation mechanisms are implemented to prevent overfitting. Additionally, the integration of the Elasticsearch, Logstash, and Kibana (ELK) stack in this layer supports detailed logging, threat hunting, and security monitoring, providing actionable real-time threat intelligence. The final segment, Layer 3, focuses on threat management and the execution of responsive strategies. This layer features intelligent Intrusion Detection and Prevention Systems (IDS/IPS), preventive solutions, event handling, and network analysis, among other strategies.

The use of open-source cyber threat intelligence (OSCTI) has grown, but extracting knowledge from unstructured OSCTI documents like security articles and reports remains challenging. THREATKG automates the gathering of OSCTI documents, extracting relevant threat information and building a detailed knowledge graph that is continually updated [29]. Central to THREATKG is a hierarchical ontology that categorizes threat data into malware, vulnerabilities,

and attack reports, allowing for structured threat representation. The processing of OSCTI involves transforming reports into a unified format, filtering out non-essential entities for Threat Knowledge Representations (TKR) using machine learning models like SVM, Random Forest, XGBoost, and LightGBM, focusing on keyword density and indicators of compromise (IOC). For threat knowledge extraction, THREATKG uses a BiLSTM model for entity recognition, along with a Piecewise Convolutional Neural Network with attention (PCNN-ATT) to identify relationships among entities, enriching the knowledge graph. A user-friendly web GUI, built with React and Elasticsearch, also allows for interactive exploration of the threat knowledge graph [29].

Integrating actionable Cyber Threat Intelligence (CTI) into Intrusion Detection Systems (IDSs) improves threat prioritization using Indicators of Compromise (IoCs) and contextual alerts for better mitigation. However, the increasing volume of cybersecurity reports complicates effective threat intelligence usage. The TriCTI system utilizes Natural Language Processing (NLP) to associate IoCs with campaign stages and extract actionable CTI from extensive reports [33]. Key phrases, identified as triggers, enhance classification by amplifying keyword significance. Targeted data augmentation addresses data scarcity, improving classification accuracy (86.99%) and F1 score (87.02%) compared to models like BERT. TriCTI processed over 29,000 reports, producing 113,543 actionable CTIs. Combining BERT pre-training and IoC feature analysis further enhances the model's efficacy. Validation against VirusTotal data confirms TriCTI's ability to cover all IoC campaign stages, aiding rapid defensive adjustments. However, challenges include the complexity of cybersecurity language and potential misclassification due to overlapping campaign stages or incorrect IoC identification.

Botnet detection using Domain Generation Algorithms (DGA) presents challenges for CTI systems that rely on blocklists, necessitating the shift to ML-based solutions, as noted by [35]. This research introduces a detection model that uses statistical features from various DGA families to improve CTI sharing and detection. Key innovations include a computable CTI model that reduces human involvement in cybersecurity decisions by integrating Explainable AI (XAI) and OSINT, enhancing output explainability through a DGA detection case study. The study also presents an efficient model for detecting DGA traffic with only seven features, outperforming previous methods, demonstrating better defenses against advanced DGA attacks (CharBot, DeepDGA, MaskDGA). Additionally, a random forest model achieves a high accuracy of 96.3% while proving more robust than several character-based deep learning models. This research validates the computable CTI paradigm, showcasing how XAI and OSINT can build trust in CTI sharing, ultimately aiding security analysts with an automated and explainable approach to OSINT analysis.

Recent advancements in cyber threat intelligence (CTI) sharing have become essential for enhancing cyber defenses through proactive collaboration to combat rising attacks. With increased organizational communication, focusing on user threat data is vital for threat identification. Establishing robust privacy policies is critical for protecting user data in the CTI sharing community. A proposed Blockchain and Federated Learning System (BFLS) [36] combines federated learning (FL) to train threat detection models without exposing organizational data, alongside blockchain technology for secure and decentralized sharing. This approach addresses privacy concerns and reduces the risks associated with centralized systems. BFLS improves the blockchain's consensus protocol to ensure that only high-quality CTIs are utilized in federated learning. The system's architecture includes a user layer for training and sharing model parameters and a blockchain layer for secure CTI exchange. Key processes involve network initialization, CTI generation, validation by a committee, aggregation through smart contracts, and committee member reselection to uphold system integrity and trust.

Cyber-threat attribution (CTA) is the process of identifying the perpetrators behind cyber-attacks. This task is complicated by attackers' use of various obfuscation and deception tactics to hide their identities. Following a cyber-incident, a digital forensic inquiry is initiated to collect evidence from network and system logs. The evidence

gathered is then shared in various report formats, including text and PDF. However, the absence of a universally accepted standard makes it difficult to extract relevant information. To address these challenges, an automatic CTA framework has been developed for the attribution and profiling of cyber threats. This framework leverages features extracted from Cyber Threat Intelligence (CTI) reports, employing natural language processing (NLP) techniques to analyze unstructured data. It also utilizes machine learning algorithms, such as random forests, for effective CTA attribution [24]. Additionally, the study introduces an embedding extraction model called Attack2vec, which is trained on domain-specific embeddings. Attack2vec extracts features related to tactics, techniques, tools, malware, targeted organizations or countries, and applications. The performance of Attack2vec is evaluated against various classification algorithms, including decision trees, random forests, and support vector machines, for the classification of cyber threat attributions.

Understanding and quantifying cyber threats is essential for enhancing cyber defense mechanisms at individual, organizational, and national levels. This study employs Convolutional Neural Networks (CNN) and Natural Language Processing (NLP) to analyze social media content related to cyber attacks and digital warfare. It introduces a method for creating a unified threat index by using keyword indexing and CNN for anomaly detection, providing a novel approach to cyber threat assessment [25]. The proposed method was validated using real-time Twitter data from October 14 to December 27, 2022, analyzing 15,983 tweets in 47 languages. This approach significantly outperformed previous monolingual analyses, resulting in the generation of 75 daily cyber threat indices for countries including China, Australia, Russia, Ukraine, Iran, and India. This information assists strategic decision-makers in refining their cyber defense strategies. However, reliance on social media data introduces certain vulnerabilities, such as potential disruptions from cyber-attacks on social media platforms and the influence of disinformation campaigns. Therefore, it is crucial to ensure robustness in the threat assessment framework.

Machine learning (ML) methods for detecting network intrusions have shown promise, particularly when trained on data from a single organization [26]. However, creating an ML-based detection system that utilizes heterogeneous data from multiple sources is challenging due to privacy concerns and differing data formats. A shared Cyber Threat Intelligence (CTI) approach enables organizations to collaboratively develop and test a robust ML-based network intrusion detection system (NIDS) while maintaining data privacy. This approach relies on a common format for network traffic data and a Federated Learning (FL) model, allowing organizations to benefit from shared intelligence without compromising sensitive information. The study presents a privacy-focused CTI method tested on two significant, non-independent NIDS datasets using a Long Short-Term Memory (LSTM) model with a 40% dropout rate. The LSTM effectively captures sequential information, reshaping the input for its requirements and passing it through three hidden layers with ReLU activation. Results indicate that this method is efficient and effective in classifying safe and unsafe traffic across organizations without the need for data exchange.

Collecting information about CTI from public sources is becoming very important to keep and reach a high-security level as systems get increasingly complex. However, these sources often have too much information. Current ML classifiers cannot handle new cybersecurity events, like new malware names or new types of attacks, because they don't

generalize well. To solve this, a continual learning approach is developed that integrates multi-level fine-tuning, GPT-3 [54] data augmentation, and ADAPET few shot learning (FSL) to enhance cyber threat detection in data-constrained environments [27]. Initially, it employs a hierarchical training strategy with the BERT model, progressively focusing from general text understanding to specific threat detection. This is complemented by GPT-3's generation of new data instances, refined through expert-led filtering to ensure relevance and quality. The ADAPET technique then adapts few-shot learning (FSL), using cloze-style questions and a verbalizer for efficient label prediction without a conventional

classification head. Results show a big improvement in the F1 score, more than 21 points higher than standard methods and more than 18 points higher than the best method in FSL [27].

Empirical evaluations have demonstrated the effectiveness of dynamic malware analysis, particularly through monitoring API call sequences in a controlled environment (sandbox). This approach serves as a robust countermeasure against code obfuscation and unidentified malware specimens. To address the urgent issue of dynamic malware detection, which is crucial for protecting computer systems from malicious software threats, a DL-based CTIMD (Deep Learning-based Cyber Threat Intelligence and Malware Detection) method was proposed by [23]. This method utilizes Convolutional Neural Networks (CNN) and Bidirectional Long Short-Term Memory (BiLSTM) networks for processing textual data. It emphasizes the analysis of program content without requiring actual execution and depends on feature extraction and signature-based matching. However, static detection methods remain vulnerable to code obfuscation and struggle to identify unknown malware. Additionally, the limitations of API call sequences in capturing the subtleties of stealthy malware behaviors—often masked as normal API calls—have led to an investigation of runtime parameters passed to these API calls.

Assessing network security situations is essential for proactive defense against threats. Analyzing this data helps understand network vulnerabilities, particularly as 5G, cloud computing, and IoT introduce new complexities. To tackle these challenges, a proposed threat forecasting model integrates blockchain technology with cyber threat intelligence (CTI) and situational awareness. This model ensures data integrity through a blockchain-based CTI framework featuring components like feeds, contributors, and miners for enhanced data sharing and analysis. The model employs linear normalization for input preparation and a Cyber Situational Awareness Engine that uses an entropy weighting k-means algorithm, supported by Adaptive Transit Search for dynamic weight adjustments [51]. It also introduces a hybrid similarity measure to select relevant alert features, which are then processed by a modified Deep Maxout Network (DMN) for intrusion prediction. Finally, based on DMN predictions, the system implements cyber attack mitigation strategies, including blacklisting, to prevent further intrusions.

CTI knowledge graphs are crucial for enhancing defense strategies for cybersecurity professionals. Constructed from CTI data organized into relation triples linking two entities by a single relation, these graphs face challenges due to the rapid increase in CTI data, which outpaces the efficacy of traditional extraction methods. This study introduces an efficient joint extraction method to extract the CTI from overcome unstructured text based on an attention-based RoBERTa-BiGRU-CRF that combines the RoBERTa [52] layer for generating word vector representations, the BiGRU layer for capturing sequential dependencies, an attention layer for focusing on relevant word dependencies, and a CRF layer for selecting the most probable tagging sequence. This model addresses traditional pipeline extraction methods' limitations by employing a joint extraction technique that simultaneously identifies and extracts entities and their relationships. It incorporates a 'BIEOS' tagging scheme for precise annotation. It facilitates the extraction process that enables the effective extraction of relation triples, significantly improving accuracy and relevance through a relation-matching technique that aligns extracted entities with suitable counterparts.

Figure 6 shows the summarized workflow of AI-based CTI methods including machine learning and deep learning based method for different downstream tasks.

3.3 LLM Methods in CTI

In the rapidly evolving field of Cyber Threat Intelligence (CTI), the use of advanced artificial intelligence (AI) techniques, particularly large language models (LLMs), presents both exciting opportunities and substantial challenges. While LLMs are highly effective at processing and analyzing vast amounts of unstructured data, they do have limitations. These models often depend on structured inputs and are primarily used to enhance the coherence of

template-driven reports, which may result in an inadequate qualitative assessment. Additionally, there are concerns regarding their ability to generate actionable recommendations and their performance in simplified or controlled environments that do not accurately reflect the complex, multi-layered situations encountered in real cyber threat scenarios. Table 5 summarizes the research papers related to LLMs in CTI, outlining their contributions, techniques, objectives, data sources, and limitations.

Table 5
Summary of papers related to the LLMs in CTI

Method	Contribution	Technique	Objective	Data Source	Limitations
AGIR [38]	Proposes a template and LLM-based framework that processes the JSON-encoded STIX graph for the automatic generation of intelligence reports	Template-based processing STIX graph and ChatGPT	Automatic CTI report generation using template and refinement using LLM	Custom STIX graphs	Dependence on Structured Input, Focuses on qualitative performance analysis lacks robust evaluation; LLM is used only for restructuring the template-based report for fluency
SecurityLLM [39]	Developing a novel encoding technique, FLLE, to transform data into a textual representation to form a template-based CTI and then leveraging the LLM to enhance the fluency of the CTI.	BERT for threat detection and FalconLLM for incident response and recovery system	Privacy-preserved multi-category classification of cyber threat for incident response and recovery system	Multi-class EdgelloTset dataset	Lacks of adapting the model to recognize new and evolving cyber threats; Constraints related to pre-trained BERT initial training data and parameters; Concerns regarding the model's ability to generate actionable and precise recommendations
Plan-act-report [40]	Proposed a decision-making framework for prompt engineering within a plan-act-report cycle for a singular action of a threat campaign, as well as a prompt chaining strategy that orchestrates the sequential decision-making sequence for a campaign involving multiple actions.	GPT-3.5-Turbo	Explores LLM's for cyber knowledge specifically, its capacity to analyze threats and advise on actions to generate intelligence regarding these threats	NMap service scan command from a Docker environment sandbox	Method limitations due to pre-trained LLM, Only a proof-of-concept within a simplified, vulnerable environment; Not accounting for the multi-hop complex scenarios.
LLM-TIKG [41]	LLM for the analysis of textualized threat intelligence to automate the identification within OSINT. Also, manual corrections on a subset of the	GPT's few-shot learning for model training. LoRA-based instruction	Threat identification and extraction of entities and their interrelations, enabling the conversion of natural language	SJTU/LLM-TIKG-dataset including thehackernews, sophos, Symantec-enterprise-blog, Fortinet, Trendmicro, and Cisa-blog	The experiment is performed only with Llama2 model; Limited evaluation strategy

Method	Contribution	Technique	Objective	Data Source	Limitations
	dataset labeled by GPT are employed in the construction of the knowledge graph.	tuning of Llama2-7B	descriptions of attack behaviors into TTPs. Facilitation of systematic assembly of a KG that integrates both low- and high-level CTI		
aCTIon [42]	Contributes a labeled dataset including 204 reports collected from renowned sources of CTI. Also, a new solution using LLM's zero-shot prompting and in-context learning capabilities which addresses some of the shortcomings of the current LLMs, hallucinations, and small context windows	GPT-3.5-turbo	Preprocessing of CTI reports to distill and refine the information into a format compatible with the constraints of the LLM. Then extraction and self-verification for the LLM, which performs the selection and categorization of the distilled information.	Developed a dataset including 204 reports spanning 12 months since February 2022	Low precision and recall; Input and output token length limitation
CRUSH [43]	Address the complexities of CTI by employing LLMs for the automated design and enrichment of EKGs.	GPT-4	Identify malicious intents and map to TIG for threat detection	Data consists of labeled script files, both malicious and clean, across various languages, validated by Microsoft's Threat Intelligence Research team, and includes blog URLs detailing threat types and TTP.	EKG construction without extensive SME involvement is novel and untested; Complex and dynamic relationships in the TIG
Threat Behavior [48]	Prompts engineered systematic exploration of prompt complexity and context. exploring the capability of GPT-3.5 in generating multiple choice questions (MCQs) and answering MCQs on cyber threat knowledge	Few-shot Prompting with GPT-3.5 and GPT-4	Evaluation of the knowledge and understanding of LLM, regarding threat behavior. Determination of the extent to which LLMs can accurately generate and answer questions related to cyber threat	MITRE ATT&CK data	A limited number of SMEs and exclusive use of multiple-choice questions from a single data source small sample size, which limits statistical confidence

Method	Contribution	Technique	Objective	Data Source	Limitations
			knowledge, including their capacity to rate the quality of the questions they generate		
ChronoCTI [44]	This study examines LLMs to identify actions and extract temporal features within textual content. Also, a dataset developed for temporal relationships among actions is compiled from ninety-four Cyber CTI reports.	GPT-3.5	Extracts temporal attack patterns from CTI reports. Identification of temporal attack patterns from historical cyberattacks	Custom dataset of temporal relations among actions from a set of 713 CTI reports	Limited dataset for experiment; Lower recall
SecurityBERT [45]	Designed an LLM that outperformed other ML and DL techniques in multi-classification of cyber attacks from IOT traffic data. Also developed is a PPFLE that encodes the IOT traffic By combining PPFLE with the BBPE tokenizer, network traffic data is transformed in a structured manner.	15-layer BERT	Attack classification from IoT traffic data	EdgelloTset cybersecurity dataset	Only a threat classification, Pretrained BERT LLM is used
LocalIntel [46]	Development of a retrieval-enhanced question-answering framework capable of producing contextualized local cyber threat intelligence by utilizing both global and local knowledge sources in response to specific queries.	GPT-3.5-turbo	retrieval and generation QA system for SoC	Trusted threat intelligence report of 326 organization-specific local Knowledge databases	Dependency on vector database, Only used for formulating defensive strategies in text format
Chatbots [47]	Evaluation of different pre-trained LLMs on attack detection and NER	ChatGPT, GPT4all, Dolly, Alpaca and Falcon	LLM for binary classification and NER.	The dataset comprised IoCs from Twitter OSINT having 31281 constructed	Struggle with domain-specific precise entity recognition due to the complexities of NER,

Method	Contribution	Technique	Objective	Data Source	Limitations
				questions and annotated for NER	Challenges when dealing with domain-specific NER, Model Limitations and Lack of Fine-Tuning
Virtual Assistant [49]	Develop virtual assistant security through three mechanisms: Reference Key, LLM Evaluator, and Self-Reminder, effectively countering various attacks.	Azure GPT-3.5 Turbo	Development of detection and defense mechanisms against various manipulation attacks targeting the system message of LLM-integrated virtual assistants.	Simulation of a real-world application scenario for Xbox customer support agent	Only an assistant that gives a textual elaboration; Can not be integrable with the mitigation system
Vulcan [53]	Developed NER and RE language models to recognize various types of cyber threats based on their semantic interconnections in unstructured CTI reports.	BERT pretrained on masked language modeling (MLM) and next sentence prediction (NSP).	Addressing the limitations of IOC-centric CTI systems in understanding and detecting rapidly evolving cyber threats because IOCs, despite their utility in identifying known threats, have a short lifespan and lack detailed technical information about threats.	CTI data from ThreatPost and Malwarebytes comprises approximately 540,000 articles where 6,791 CTI data instances and 4,323 relationships are annotated for 6 and 7 types of CTI data and relationships respectively.	Vulcan can Identify a fixed set of CTI data types and their semantic relationship that limits its adaptability to new or evolving cyber threat vectors that may not fit within the predefined categories; Vulcan's language models for extraction are trained on a specific set of entities and relationships, which may not fully encompass the diversity of cyber threats.

The importance of CTI reporting in today's risk management is very pivotal. As the number of CTI reports grows, there is a clear need for automated systems to make the creation of these reports easier and more efficient. Even though NLP techniques have improved and can handle text data, they still need help dealing with the complexity of different data sources and how they are connected. AGIR [38] was proposed to help security analysts by making it easier to create detailed intelligence reports from structured entity graphs. AGIR is based on a dual-module approach: the template-based and neural-based modules. Initially, the template-based module accepts a JSON-formatted STIX graph and a user-defined report type to select an appropriate template. This selection process involves categorizing the input into one of four report types, Overview, Subject, Timeline, or Vulnerability Report, each handling different aspects of threat intelligence. The module then extracts entities and their relationships from the STIX graph, utilizing the Knowledge Base (KB) to enrich the report with detailed information on each entity, including its historical data and associated intelligence, such as IOCs and TTP. To enhance the readability and fluidity of the reports generated by the

template-based module, AGIR incorporates a neural-based module leveraging ChatGPT that refines the initial report to ensure its fluency while maintaining the integrity of the information [38].

Leveraging LLM for network-based threat classification, incident response, and recovery systems, SecurityLLM [39] integrates two key components: SecurityBERT for cyber threat detection and FalconLLM [54] for incident response and recovery. SecurityBERT initiates the process by collecting and analyzing cybersecurity data from various sources. It extracts relevant features from network traffic logs and transforms them into a textual representation using a novel encoder called Fixed-Length Language Encoding (FLLE), combined with the Byte-Level BytePair Encoding (BBPE) tokenizer. This transformation allows SecurityBERT to effectively process and analyze the data, accurately detecting a wide range of cyber threats. FalconLLM takes over once a threat is detected for the incident response and recovery system. It utilizes its advanced language understanding capabilities to analyze the threat, propose mitigation strategies, and guide the recovery process. The performance of SecurityLLM significantly surpasses traditional ML and DL methods, showcasing the potential of LLMs in enhancing cybersecurity measures.

Enhancing cybersecurity defenses by integrating LLMs, GPT-3.5-Turbo [40] is used for threat identification and mitigation strategies. This approach is motivated by the wide range of threat actor capabilities, from novices using basic scripts to identify vulnerabilities to sophisticated APTs leveraging zero-day exploits for stealthy operations. The goal is to augment defensive systems by tapping into the LLM's ability to reason about cyber threats and suggest actionable countermeasures. The method progresses through an exploratory phase where the LLM's ability to understand cybersecurity queries and generate executable commands is assessed, demonstrating a deep understanding of cyber operations. The system design adopts a three-stage campaign model of reconnaissance, exploitation, and exfiltration, guided by a decision-making process akin to that of human threat actors. This involves tactic selection based on the campaign's state, execution of commands for the selected tactic, and output translation to evaluate the action's success and inform subsequent steps. An automated agent employing prompt engineering and a finite state machine is developed to navigate these stages autonomously. Experiments within a Docker-based sandbox environment validate the feasibility of using LLM recommendations against pre-configured vulnerable targets, assessing the LLM's cybersecurity knowledge and capability to generate actionable outputs [40].

Often unstructured, OSINT presents challenges for direct application in subsequent detection and defense mechanisms. The construction of a knowledge graph utilizing OSINT enhances the application of such information for intrusion detection purposes. LLM-TIKG [41] employs an LLM, Llama2-7B for the development of a knowledge graph (KG) derived from unstructured OSINT for topic classification, entity and Relationship Extraction, and TTP Classification. This method capitalizes on the advanced few-shot learning [54] capabilities of the GPT model for data annotation and augmentation, facilitating the creation of the required dataset. Subsequently, the Llama2-7B model undergoes finetuning

via a LoRA-based [41] instruction tuning approach, empowering it to classify input texts, extract entities and relationships pertinent to threat intelligence, and delineate TTPs from descriptions of attacks. Utilizing this refined model, it conducts topic classification of the gathered reports, identifies entities and their relationships, and extracts TTPs from the descriptions of attacks.

CTI equips security operators with essential data to combat cyber threats and respond to attacks. The majority of CTI remains unstructured, such as in text reports, and this unstructured CTI, detailing attackers, victims, tools, and attack patterns, is crucial for enabling threat-hunting activities. aCTIon [42] leverages the zeroshot prompting and in-context learning capabilities of LLMs like GPT3.5 through two custom information extraction pipelines, aiming to improve upon the limitations of existing automated tools for CTI extraction. This solution introduces a two-step querying process designed to mitigate issues like hallucinations and limited context windows, characteristic of current LLMs. It

involves preprocessing the input report to distill information within the constraints of the target LLM, followed by the definition of extraction and selfverification prompts for the LLM. This allows for the selection and classification of relevant information extracted from the unstructured text. The results demonstrate that aCTIon significantly outperforms previous methods across various tasks, improving the F1-score by 10 to 50 percentage points. Alongside aCTIon, it introduces a large, open benchmark, a labeled dataset of 204 CTI reports from renowned sources and their STIX bundles, averaging 2133 words per report and peaking at 6446 words. Security analysts classified these reports using the MITRE ATT&CK Matrix, encompassing over 190 detailed entries, to ensure accurate classification of attack patterns [42].

Enterprise Knowledge Graphs (EKGs) serve as potent instruments for the representation and inference within intricate and evolving domains, including CTI. Nonetheless, the design and development of these graphs present significant challenges, particularly in the context of managing data sources that are heterogeneous and characterized by noise.

CRUSH [43] uses LLMs for designing and populating EKGs, specifically focusing on a Threat Intelligence Graph (TIG). Utilizing GPT3.5, GPT4, and ChatGPT, CRUSH automates the extraction, inference, validation, and summarization of information from diverse sources, like threat reports and scripts, to identify malicious intents and correlate them with cyber threat intelligence. It involves extracting relevant entities, relationships, and properties to populate the TIG dynamically without extensive subject matter expert (SME) involvement. It addresses common LLM challenges such as hallucination and limited reasoning through techniques like Retrieval Augmented Generation (RAG), search augmentation, and planning frameworks.

Enhancing cyber defense requires a deep understanding of adversary behaviors and effective mitigation strategies, which necessitates expertise from subject matter experts (SMEs). The proposed LLM-based MITRE ATT&CK Framework, as outlined by [48], examines the proficiency of GPT-3.5 in recognizing threat behaviors. This framework is a widely recognized collection of adversarial tactics and techniques derived from empirical observations. The study tests various prompts using the LLM, starting with GPT-3.5 for question creation. It first generates questions focused on threat behavior mitigations from the ATT&CK Framework, using prompts with and without additional contextual information. These prompts are designed to ensure that the output (i.e., the questions) aligns with the complexity typically found in the US Bar exam. The process involves creating multiple-choice questions on cyber threat behaviors and asking SMEs to evaluate the accuracy and clarity of these questions. Following this, GPT-3.5 is utilized to respond to the generated questions, employing diverse prompts to extract accurate answers regarding the threat behaviors. The assessment of cyber knowledge within the MITRE ATT&CK Framework consists of two main steps: (1) Constructing the assessment, which involves designing prompts that elicit clear and correct questions and answers; and (2) Implementing the assessment, where both SMEs and LLMs are used to respond to the formulated questions. Specifically, the study focuses on the effectiveness of different prompting strategies for LLMs in generating multiple-choice questions related to cyber threat knowledge found in the MITRE ATT&CK Framework.

Security professionals are tasked with identifying patterns of hostile behavior to develop proactive defense strategies. Cyber Threat Intelligence (CTI) reports are a valuable resource for these experts, as they provide detailed accounts of historical cyberattacks. These reports include nuanced information and insights related to the sequence and timing of actions, known as the temporal relationships among these actions. By extracting these patterns from the recorded actions, security professionals can automate the delivery of organized and practical insights into the adversary's modus operandi during past cyber incidents. ChronoCTI, developed by [44], is a systematic framework designed to automate the extraction of temporal attack patterns from CTI reports on previous cyber incidents using GPT-3.5. ChronoCTI involves creating a definitive dataset of temporal attack patterns and integrating cutting-edge advancements in large language models (LLMs). When applied to a collection of 713 CTI reports, ChronoCTI

identified 124 distinct temporal attack patterns, which were further classified into nine specific categories. While ChronoCTI excels in precision, it is noted to have limitations in recall.

As the number and variety of cybersecurity threats continue to increase, promptly identifying and addressing these threats has become increasingly important. The rapid growth of IoT (Internet of Things) devices has further heightened the demand for effective methods to accurately detect network threats in IoT environments without consuming excessive computing power. SecurityBERT, developed by [45], utilizes the BERT model to identify cybersecurity threats in IoT networks. In SecurityBERT, Privacy-Preserving Fixed-Length Encoding (PPFLE) is combined with the BBPE Tokenizer to efficiently organize network traffic data. This combination allows the transformation of text traffic data into a structured format that is ready for further analysis. The final step in SecurityBERT involves classifying the attacks, which enhances the accuracy of systems designed to detect intrusions. Studies indicate that SecurityBERT outperforms traditional machine learning (ML) and deep learning (DL) techniques, such as CNNs (Convolutional Neural Networks) and RNNs (Recurrent Neural Networks). In fact, SecurityBERT achieved an impressive accuracy of 98.2% in identifying fourteen different types of attacks, surpassing previous methods, including GAN-Transformer architectures and CNN-LSTM models. With a quick analysis time of less than 0.15 seconds on a standard CPU and a compact size of just 16.7MB, SecurityBERT is exceptionally well-suited for analyzing real-world traffic and can be effectively used on IoT devices with limited resources.

Security operations center (SoC) analysts collect threat intelligence from publicly available global threat databases and tailor this information to meet their organizations' specific requirements. These professionals rely on internal databases that serve as repositories of organization-specific knowledge, containing reliable cyber intelligence, essential operational insights, and relevant data. The process of synthesizing global and local knowledge into unique threat response and mitigation strategies for an organization is notably labor-intensive. The GPT-3.5-based LOCALINTEL system, as described by [46], operates by retrieving threat intelligence from global databases and then using its local knowledge database to contextualize this information specifically for an organization. This system is structured around three primary phases: acquiring global threat intelligence, extracting relevant local knowledge, and generating contextualized outcomes. First, it gathers intelligence from global databases. Next, it extracts pertinent information from the organization's local knowledge base. Finally, these streams of knowledge are integrated using a generative process to produce contextualized intelligence tailored to the organization's specific needs.

The exchange of knowledge about emerging threats is crucial and serves as the foundation of Cyber Threat Intelligence (CTI). This research by [47] explores the potential of chatbots, including ChatGPT, GPT4all, Dolly, Alpaca-LoRA, and Falcon, to identify cybersecurity-related discussions within Open Source Intelligence (OSINT). The study evaluates the proficiency of current chatbot models in performing Natural Language Processing (NLP) tasks, focusing on binary classification and Named Entity Recognition (NER) using a dataset gathered from Twitter. In binary classification tasks, the commercial model ChatGPT (GPT-4) achieved an F1-score of 0.94, while the open-source model GPT4all recorded an F1-score of 0.90. However, when it comes to recognizing cybersecurity entities, the effectiveness of chatbot models is significantly diminished, indicating their limitations. This investigation reveals the ability of large language model (LLM)-based chatbots to identify cybersecurity-related tweets from a Twitter dataset, demonstrating their adaptability to new and uncharted contexts, even in the absence of direct training data. In terms of NER tasks, important findings emerged from experiments with chatbot models. Primarily, the ability of current chatbots to detect cybersecurity entities faces considerable limitations. Despite being trained on extensive datasets, these models did not achieve the expected performance on our test data, highlighting the need for further focused research and refinement in the application of chatbot models for cybersecurity entity identification. While there have been significant advances in various areas, ChatGPT, although not as precise as specialized NER models and exhibiting notable deficiencies, still holds potential as an initial tool for text pre-annotation [47].

The integration of large language models (LLMs) with virtual assistants has led to significant changes in communication methods. However, it is essential to recognize that over-reliance on this technology can make these systems vulnerable to exploitation by malicious actors through the use of carefully crafted prompts. This type of exploitation poses a serious risk, undermining the accuracy and reliability of virtual assistants' responses. To address these concerns, it is necessary to implement detection and defense strategies to protect the security and integrity of virtual assistants. The proposed framework evaluates three strategies aimed at mitigating attacks directed at the system message: the integration of a reference key, the use of an LLM evaluator, and the adoption of a Self-Reminder technique [49]. The findings from this investigation are highly relevant to the development and deployment of LLM-integrated virtual assistants across various sectors, including customer support, healthcare, and finance. This underscores the importance of integrating detection and defense strategies to protect these systems from malicious interventions while ensuring their proper functionality.

Research on evolving cyber threats has primarily centered on developing Cyber Threat Intelligence (CTI) systems that collect open-source data, focusing mainly on Indicators of Compromise (IoCs) such as file hashes and IP addresses. However, the ability of IoC-focused CTI systems to understand and identify threats is limited for two main reasons. First, IoCs serve as evidence of a breach but do not provide detailed technical information about the underlying threats. Second, attackers frequently change their infrastructure and static IoCs, resulting in these indicators having a very short period of relevance. To address the limitations of traditional IoC data, [53] have automated the extraction and analysis of CTI from unstructured text sources. This approach uncovers a diverse array of CTI data that goes beyond conventional IoCs, helping to reveal the technical foundations and semantic relationships of cyber threats. Vulcan leverages advanced neural language models and fine-tunes both Named Entity Recognition (NER) and Relation Extraction (RE) models. At its core, Vulcan integrates a graph database to store the extracted CTI data, providing search APIs for security analysts. The architecture of Vulcan consists of several key components: a data scraper and pre-processor for data collection, a Threat Entity Identifier (TEI), a Threat Relation Identifier (TRI) for data extraction, and an Entity Linker for data refinement.

Figure 7 illustrates the workflow of large language model (LLM)-driven Cyber Threat Intelligence (CTI) approaches and the general process of multi-purpose LLM-based CTI methods. Heterogenous data sources can be utilized to create a comprehensive data corpus, and any LLM model can be trained using one of the specified training paradigms for pre-training and downstream tasks.

DISCUSSION

The development of a holistic CTI framework for threat detection and forecasting is very essential for organizations to efficiently and accurately detect and mitigate prospective risks. As cyber threats gain in sophistication, surpassing the efficacy of conventional security measures, a proactive CTI framework becomes indispensable. It facilitates the early identification of vulnerabilities within an organization's infrastructure, the detection of irregular patterns suggestive of cyber misconduct, and the systematic distribution of threat intelligence to relevant stakeholders. Hence, this discourse advocates for implementing a CTI framework designed to augment the cyber threat response capabilities of business entities.

This study introduces different state-of-the-art CTI methods for the detection and forecasting of cyber-attacks, enabling early and effective communication with potential threats and relevant security actions to prepare for future challenges. Figure 8 shows a summary of the reviewed studies.

Such intelligent methods prevent cyber intrusions by providing extra time to create better defensive strategies/tools in competitive cyber environments. Proactive CTI methods also help reduce uncertainty in prioritizing current security

measures or introducing new security solutions. Cyber-security organizations should prevent the most rapidly growing attacks identified in the early stages of the Threat to Capability (TTC) cycle or attacks within fast-growing or emerging trends on future predictions. Zero-day attacks exploit vulnerabilities unknown to developers before issuing a corrective patch or solution. These exploits pose a significant threat, capable of compromising even the most strengthened systems, remaining undetected for extended periods. Consequently, they can cause considerable harm to an organization's reputation, financial stability, and consumer confidence. ML in addressing zero-day exploits, offering a more proactive CTI method trained on preprocessed structure data. Utilizing CNNs to examine complex, high-dimensional datasets, CTI methods provide entities with the foresight to thwart zero-day exploits before their inception, addressing a challenge for which a solution is currently nonexistent despite detection capabilities.

This literature review aims to outline the current leading edge and future research directions for CTI distribution and its characteristics. The review was executed by exploring journal databases, academic libraries, and scholarly search engines. The issues explored in this review originate from challenges that the proposed approach encounters technically in the current scenario and real problems that cyber entities face today. This review aims to present a comprehensive overview of the various challenges associated with cyber-threat detection, identification, recognition, and mitigation. Several scholars have proactively tackled stages of CTI sharing that remain largely unexplored by the broader practitioner community. Fundamental obstacles remain, including the basic implementation of a CTI mechanism and persuading decision-makers to allocate resources towards it. Moreover, integrating monitoring and detection tools with threat intelligence platforms presents its own set of challenges. Upon initiating the consumption and exchange of intelligence, stakeholders often find themselves overwhelmed with information, considering its relevance. Various providers offer threat intelligence platforms as an initial entry into the field of CTI through the utilization of crowdsourced intelligence from CTI databases. Current research primarily focuses on strategies for encouraging long-term collaborations among decentralized entities and streamlining sharing mechanisms. It is acknowledged that CTI sharing has captured the interest of numerous organizations aiming to adopt more proactive CTI methods rather than a purely reactive one.

In non-learner CTI methods, template-based, rule-based, graphical, and statistical methods highlight a diverse range of approaches aimed at element extraction, assessment, detection, recognition, and sharing of CTI and utilize various NLP techniques to extract relevant information from CTI reports. TIAM [13] employs vector space models and regular expressions to analyze unstructured CTI reports, aiming to classify and identify cyber threats based on textual features. This method, while innovative in the quality assessment of CTI, faces obstacles, including data sparsity and the absence of real-time threat evaluation capabilities. On the other hand, DFR [14], ABC [15], Decentralized CTI [21], and BLOCIS [22] leverage the potential of digital forensic readiness and blockchain technologies to enhance cybersecurity measures. These studies address the augmentation of cybersecurity defense mechanisms but also reveal critical issues such as the risk of false positives, implementation complexity, and the challenges associated with blockchain's scalability and reliability.

Further, studies like *CyRiPred* [16] and AttackKG [18] utilize statistical methods and technique templates for cyber risk identification and the construction of attack graphs. These approaches are commended for their structured analysis but critiqued for their limited adaptability to the dynamic nature of cyber threats. Comprehensive CTI frameworks, Eight-step CTI [17] and SCERM [50], offer end-to-end solutions for CTI management, including collection, analysis, refinement, and sharing. However, these frameworks face challenges such as the absence of real-time intelligence and susceptibility to misinformation from unverified sources. Lastly, the introduction of semantic schemas and lifecycle frameworks, SECDFAN [19] and Cyber Defenders [20] represents an effort to standardize CTI sharing and optimize cyber defense mechanisms within SOCs. Despite their innovative approaches, these frameworks encounter difficulties in handling the dynamic complexities of cyber threats, reliance on NLP techniques,

and the assumption of a standardized threat landscape that may not reflect the specific and varied threats faced by different organizations. Collectively, these studies articulate the imperative need for advanced, adaptive CTI frameworks that not only address the dynamic cyber threat landscape but also ensure the integrity, reliability, and timeliness of threat intelligence. They highlight the evolving nature of cyber threats and the corresponding requirement for scalable CTI solutions that provide real-time insights into potential risks, especially in zero-day attacks.

Data-driven AI-based learnable CTI methods enhance cyber threat detection, forecasting analysis, and information sharing. One notable area involves the detection of malware and the attribution of threat actors. Studies such as CTIMD [23] and CTI attribution [24] delve into dynamic malware detection and the profiling of cyber-threat actors, respectively. The former leverages TextCNN and BiLSTM for analyzing API call sequences to detect unknown malware, while the latter utilizes an Attack2vect-based framework alongside Random Forest for accurate threat actor identification. These studies have certain limitations like the potential for intrusion through concealment techniques and the challenges in processing unstructured data for actionable insights. Another significant area explored is the utilization of social media analytics for threat detection. Social-CNN [25] and TIMiner [31] employ CNNs and word embeddings to extract and categorize CTI from social media content. This approach demonstrates the utility of public data sources in generating threat indices and identifying IoCs. However, the reliability of social media data introduces vulnerabilities, particularly to misinformation and the inadequacy in detecting sophisticated polymorphic threats. Further, the literature highlights the development of privacy-preserving techniques employing federated learning. Studies like Intrusion Detection [26] and BFLS [36] propose innovative schemes that facilitate collaborative cyber attack detection while safeguarding data privacy through federated learning. These methods emphasize the critical role of shared CTI formats and blockchain technology in achieving secure, decentralized CTI exchanges. Despite their promise, these approaches face challenges, including inefficiencies in processing diverse data sets and potential security loopholes in consensus protocols. Lastly, the construction of knowledge graphs for improved CTI sharing is addressed by studies such as THREATKG [29] and CyberEntRel [52]. These works focus on aggregating and structuring CTI into accessible knowledge graphs, utilizing ML, NLP, and novel joint extraction techniques to enhance the organization and sharing of CTI. While these methods mark significant progress in CTI management, they also recognize existing limitations, including the adversarial nature of open-source CTI sources and the complexity of extracting overlapping relational triples.

The integration of LLMs into CTI forecasting signifies a transformative shift across various dimensions of cybersecurity practices. Studies demonstrate the diverse applications of LLMs, ranging from automating intelligence report generation to enhancing threat detection and from constructing detailed knowledge graphs to improving the overall understanding and classification of cyber threats. One notable application is the automation of report generation, as seen with AGIR [38], which leverages a combination of template-based frameworks and LLMs like ChatGPT for producing intelligence reports from structured inputs. This method exemplifies how LLMs can streamline complex processes, albeit highlighting a reliance on structured data and a qualitative approach to performance analysis. In threat detection and incident response, studies such as SecurityLLM [39] and SecurityBERT [45] utilize BERT-based models to accurately detect and classify cyber threats, particularly in IoT environments. These models emphasize the importance of privacy-preserving mechanisms and innovative data transformation techniques, showcasing the LLM's efficiency in navigating and mitigating cybersecurity threats across diverse data sets.

LLMs further enhance the construction of knowledge graphs and the analysis of entity relationships through projects like LLM-TIKG [41] and CRUSH [43]. These efforts focus on deriving structured intelligence from unstructured open-source data, underscoring the LLM's capability to distill complex data into actionable intelligence, thereby facilitating a more organized approach to CTI application. Moreover, the enhancement of cybersecurity discourse understanding

through initiatives such as aCTIon [42], Chatbots [47], and Threat Behavior [48] reflects the versatility of LLMs in processing CTI reports, performing binary classification and NER, and generating content on cyber threat behaviors. Despite facing challenges in domain-specific entity recognition and providing precise actionable insights, these studies illustrate the potential of LLMs in broadening the comprehension of cyber threats. Additionally, the extraction of temporal attack patterns and the production of contextualized intelligence, as demonstrated by ChronoCTI [44] and LocalIntel [46], highlight the innovative use of LLMs in deciphering the temporal and contextual nuances of cyber threats. This facilitates the development of proactive defense strategies, offering a deeper insight into the adversary's modus operandi. Lastly, the focus on virtual assistant security and the advancement of cyber defense mechanisms, as seen with Virtual Assistant [49] and Vulcan [53], address the challenges of securing LLM-integrated platforms and enhancing threat detection capabilities. These contributions signify the crucial role of LLMs in securing interactive systems and providing a comprehensive understanding of cyber threats beyond traditional IoCs.

Collectively, the integration of LLMs in CTI forecasting marks the beginning of a new era in cybersecurity, where the automation of intelligence reports, the enhancement of threat detection, the construction of knowledge graphs, and the nuanced analysis of cyber threats converge to boost the proactive defense against evolving cyber threats. This paradigm shift highlights the potential of LLMs in revolutionizing cybersecurity practices because of the long-term understanding of unstructured CTI reports, offering novel approaches to threat intelligence analysis and application.

CHALLENGES AND FUTURE RESEARCH DIRECTIONS

This systematic literature review provides valuable insights into the advancements in CTI forecasting through the application of AI and LLMs, it is important to recognize its limitations. Future research should aim to address these gaps, focusing on the integration challenges of LLMs, the ethical implications of AI in cybersecurity, and the continuous monitoring of emerging trends and technologies in this dynamic field. The limitations of this systematic literature review on advancing CTI forecasting with AI and LLMs can be summarized across various dimensions:

1. Detection and classification of network intrusions requires an understanding of the nature of the attacks and their respective categories. CTI methods, formulated to address cybersecurity challenges, ought not to be narrowly concentrated on singular issues, such as malware detection. Instead, a holistic approach should be adopted, integrating different AI models with a variety of ML, DL, and LLM techniques and encryption algorithms to cover an extensive range of attack vectors comprehensively. In the future, LLM-based methods will be key techniques because of the ability of a single LLM model to do multiple downstream.
2. The effectiveness of cybersecurity applications is frequently impacted by the ratio of normal to malicious data present in the training dataset. To assess the efficiency of a developed cybersecurity application, it is critical to analyze performance metrics, including processing speed, resistance to data poisoning, imbalance data handling, the True Positive Rate, and the False Positive Rate. Such evaluation parameters are indispensable for determining the application's ability to identify and respond to cyber threats accurately.
3. Acquiring authentic Cyber-threat datasets in real-time presents a significant challenge within the field of CTI. Typically, the available Cyber-threat datasets are mostly structured and constructed through experimental methods, while most CTI reports are unstructured. Therefore, future research endeavors should concentrate on exploring various open-source datasets and benchmark models. This approach will enhance cybersecurity systems' understanding and detection capabilities by providing more diverse and realistic data for analysis. DL and LLM-based CTI report analysis require high resource requirements including high-performance GPUs, ample storage, and computational power. Furthermore, analytics of extensive resources demand highly scalable algorithms, consume minimal power, exhibit flexibility, and optimize local bandwidth usage. Additionally, many

existing designs are characterized by their significant computational costs, intractability, and complex hyperparameter configurations.

4. The application of DL and LLMs in CTI is associated with considerable cost complexity, particularly when addressing errors. This complexity arises because DL and LLM models often operate as "black boxes," making it challenging to pinpoint the precise origin of errors. Moving forward, conducting thorough analyses of the root causes behind cyber attacks is imperative. Such detailed investigations will facilitate the development of an active learning framework tailored to CTI applications, enabling more effective and efficient error resolution and threat mitigation.
5. Despite a comprehensive review, the literature scope is inherently limited by the publication only focused on CTI, rather broader area of cyber security. Moreover, LLM has very recently emerged and comprehensive evaluation is required to determine the accuracy of LLM-based CTI methods.
6. While the review covers a broad range of applications of AI and LLMs in CTI, the depth of analysis on each specific application may be limited. Detailed technical evaluations or comparative analyses of different models and approaches are beyond the scope of this literature review, which may limit the utility for researchers seeking an in-depth understanding of specific methodologies.
7. The literature review may focus more on successful applications of AI and LLMs in CTI forecasting, potentially underrepresenting studies that discuss these technologies' limitations, failures, or challenges. This could provide a skewed optimism towards the efficacy of AI and LLMs in CTI without fully addressing their limitations.
8. The review discusses the pioneering application of LLMs in automating intelligence report generation, enhancing threat detection accuracy, and constructing comprehensive knowledge graphs. However, it may not fully address the practical challenges in integrating these models into existing cybersecurity frameworks, including issues related to data privacy, model interpretability, and the dynamic nature of cyber threats.
9. While the review mentions data privacy concerns and the dynamic nature of cyber threats, it may not delve deeply into the ethical considerations of using AI and LLMs in CTI. The implications of automated data collection, analysis, and sharing on individual privacy rights and data security are critical areas that require further exploration.
10. The findings and conclusions drawn from the reviewed literature may not be universally applicable across different cybersecurity contexts, industries, or threat landscapes. The effectiveness of AI and LLM applications in CTI forecasting can vary significantly based on specific datasets, use cases, threat models, and organizational capabilities.

CONCLUSION

This SLR thoroughly examines CTI, highlighting a crucial change in cybersecurity. This change is driven by the fast-paced development of cyber threats and the need for more advanced detection and forecasting methods. Through an in-depth review that includes traditional CTI methods, the use of AI, and the innovative application of LLMs, this study sheds light on the varied strategies for enhancing cyber security measures and preventing cyber attacks. Traditional CTI methods have been foundational but show limitations against modern cyber threats' complex and evolving nature. The reliance on outdated, signature-based security is proving to be less effective, pointing to the necessity for security measures that are both adaptive and anticipatory. The introduction of AI into CTI forecasting marks notable progress, providing improved ways for detecting malware, identifying threat actors, and leveraging social media for intelligence gathering. Despite these advancements, challenges such as concerns over data privacy, the need for common data formats, and the susceptibility to misinformation and sophisticated evasion tactics remain. The integration of LLMs into CTI forecasting signifies a major advancement, promising a new direction in cybersecurity practices. LLMs facilitate the automatic creation of intelligence reports, enhance the accuracy of threat detection,

help in building detailed knowledge graphs, and improve the understanding and classification of cyber threats. These models not only simplify complex processes but also provide deeper insights into adversary tactics and strategies, leading to more proactive and informed defense strategies. Yet, the path forward is filled with challenges. The ever-changing nature of cyber threats, the short-lived relevance of traditional IoCs, and the difficulties in integrating LLMs into current cybersecurity frameworks are significant obstacles. Additionally, the potential for LLMs to generate misleading information and the ethical issues related to privacy in data sharing require careful consideration.

Declarations

Author Contribution

Hilalah Alturkistani authored the main manuscript, while Suriayati Chuprat reviewed and validated it.

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Figures

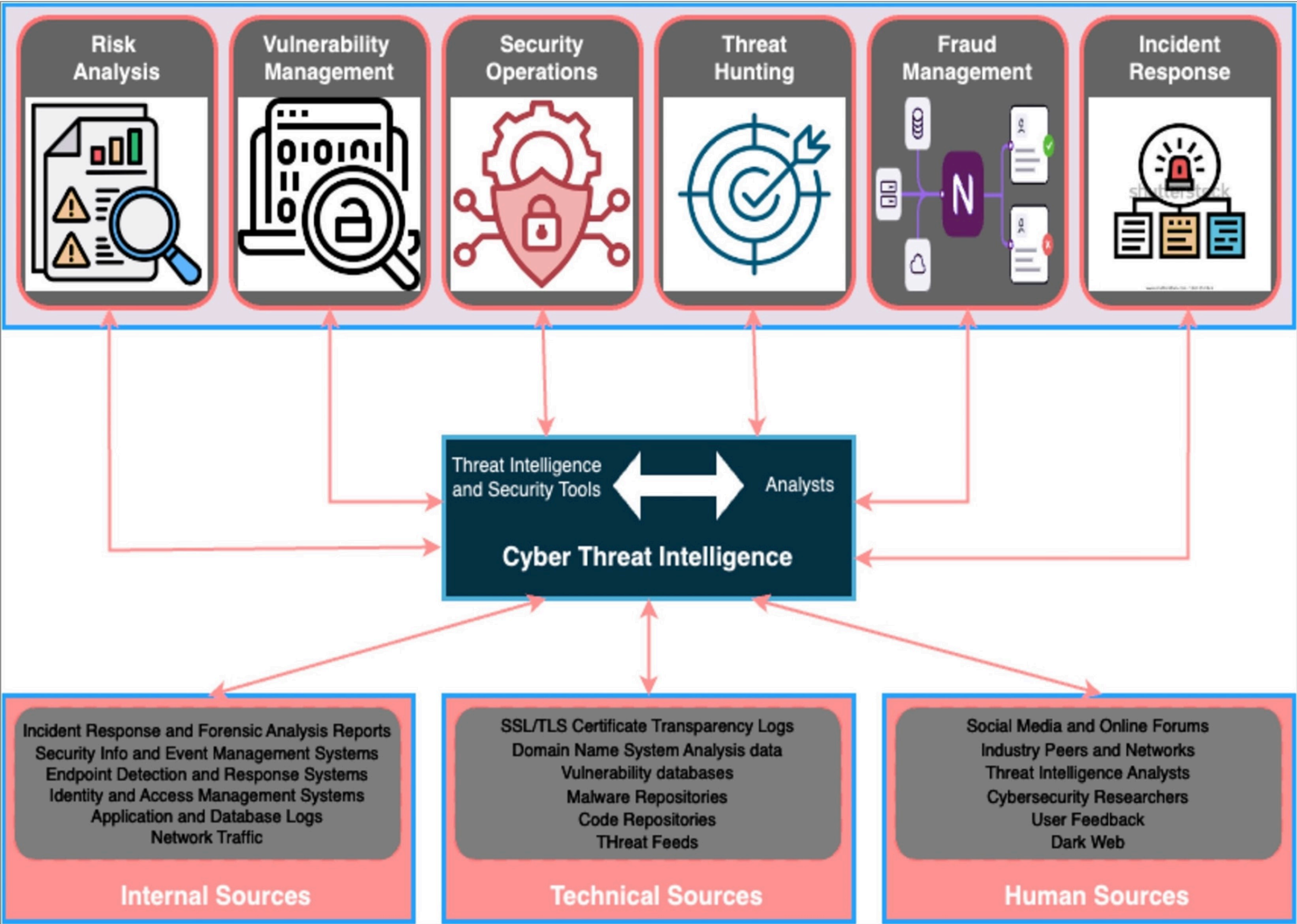


Figure 1

General Workflow comprises a six-phase process of cyber threat intelligence

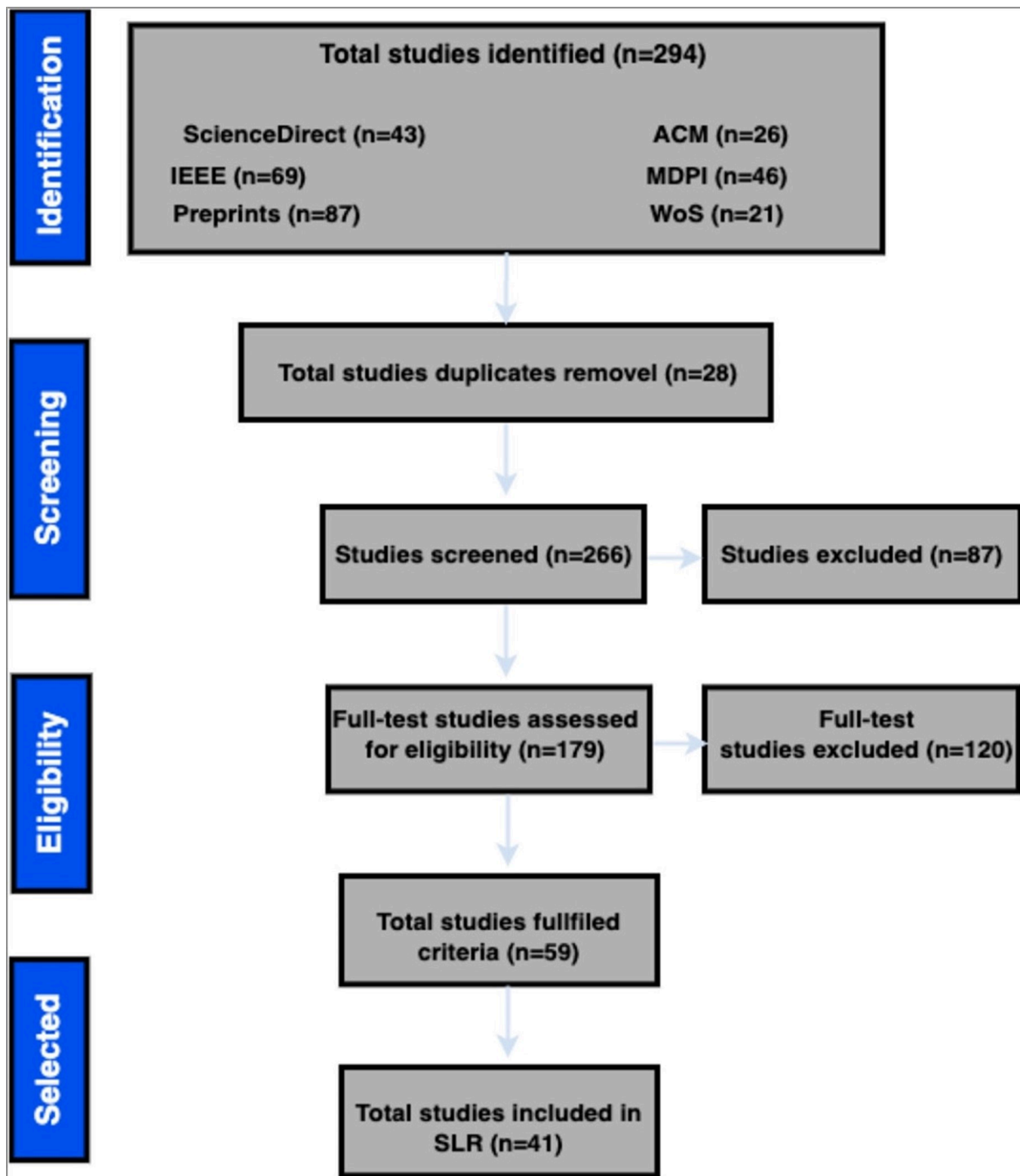


Figure 2

SLR workflow for study selection

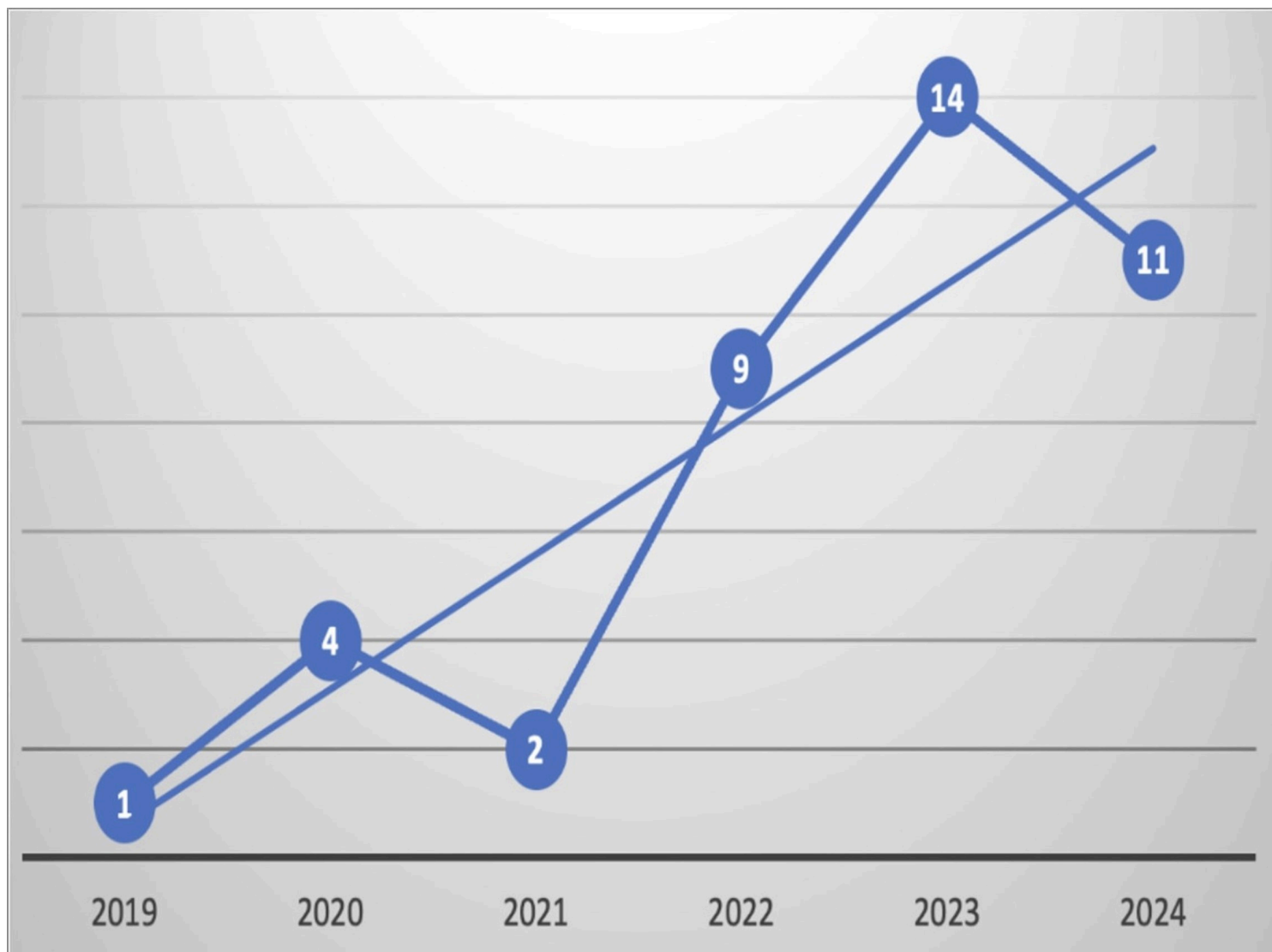


Figure 3

Number of research studies on CTI published from 2019 to February 2024

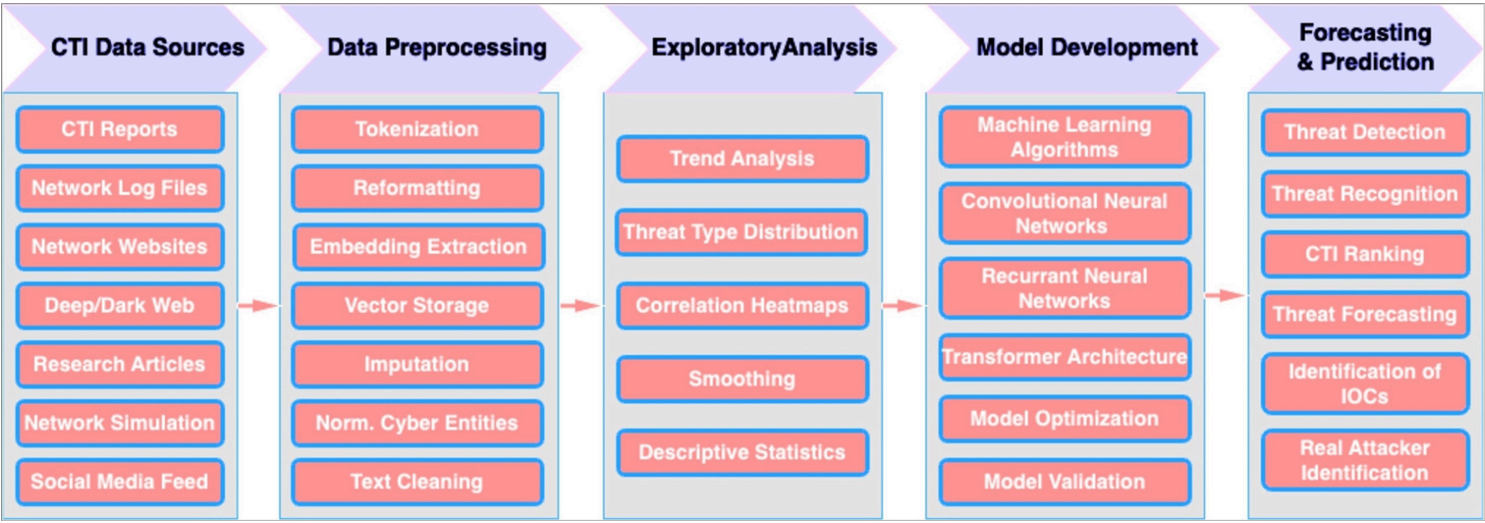


Figure 6

Workflow of AI-based CTI methods

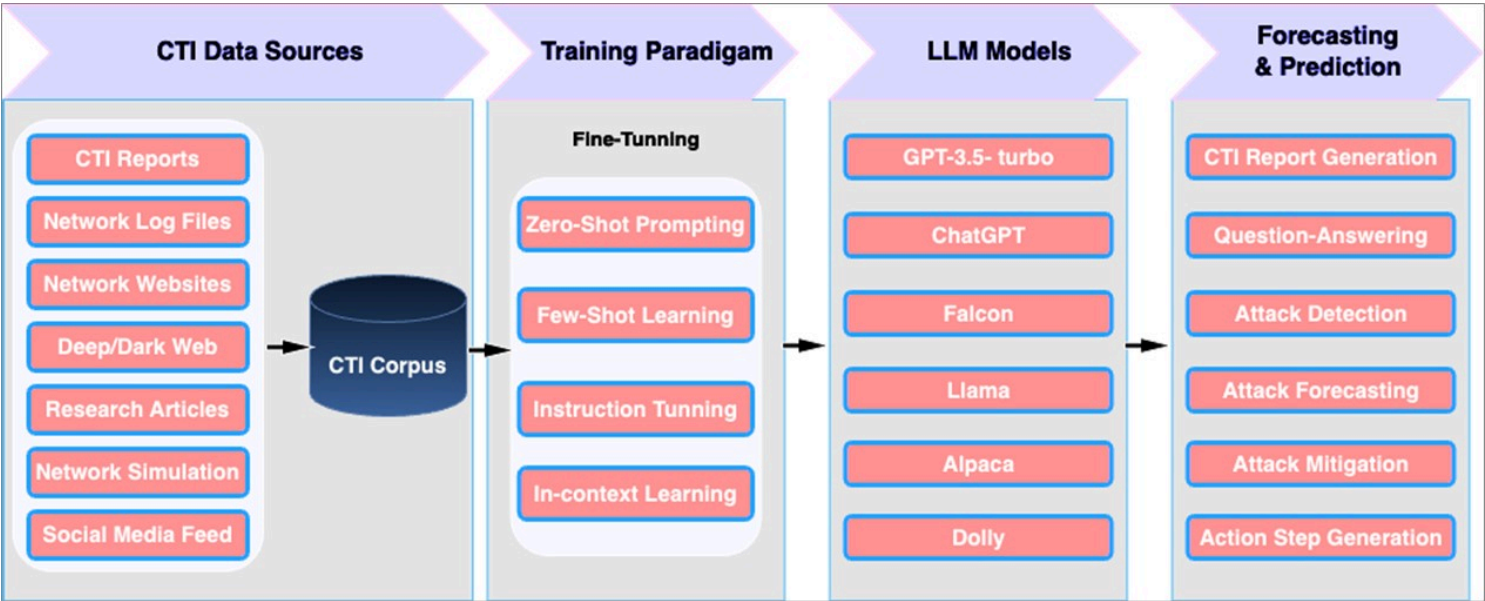


Figure 7

General workflow of multi-purpose LLM-based CTI methods

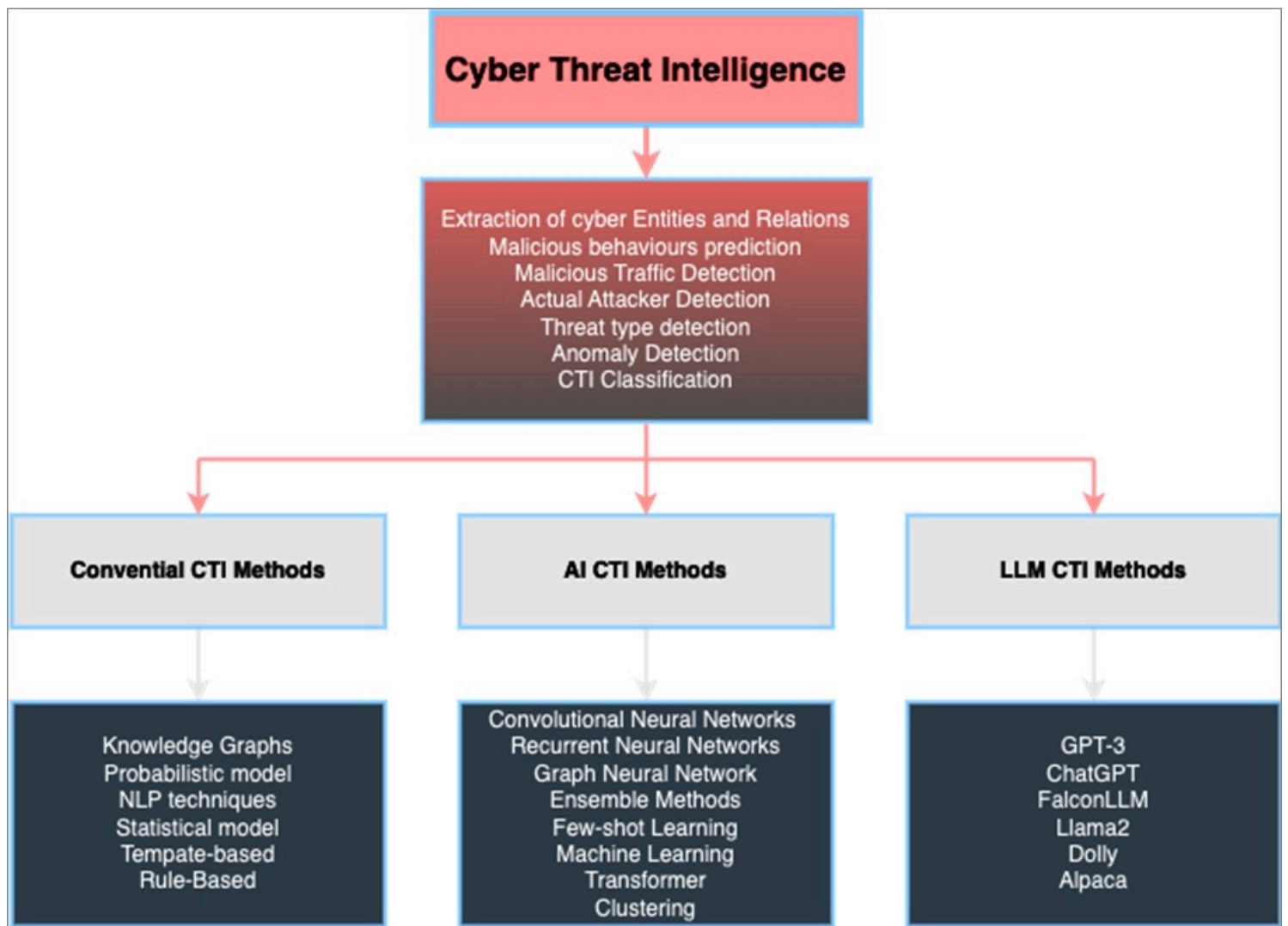


Figure 8

Summary of Literature that is reviewed in this SLR